

FORESEE F35SQB004G 4Gbit SPI NAND Flash 数据手册

中文工程翻译版

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【FORESEE F35SQB004G 4Gbit SPI NAND Flash 数据手册】

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FORESEE®

4Gbit SPI NAND Flash

F35SQB004G

Datasheet

LM-00130

Rev 1.0

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Revision History

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【1 概述】

F35SQB004G 是一款 4Gbit (512M x 8bit) 串行 NAND Flash, 采用单路 3.3V VCC 供电。

器件支持标准 SPI、Dual SPI 和 Quad SPI, 相关信号包括串行时钟 CLK、片选 CS#、串行数据 SIO0 (DI)、SIO1 (DO)、SIO2 (WP#) 和 SIO3 (HOLD#)。SPI 时钟最高支持 166MHz。

F35SQB004G 支持 JEDEC 标准厂商/器件 ID、唯一 ID (Unique ID)、1 个参数页以及 62 个 OTP 页。芯片内部集成 8-bit ECC 逻辑, 出厂默认使能, 也可通过命令关闭或重新使能。

【2 特性】

供电: VCC = 2.7V~3.6V。

组织结构:

- 内部 ECC 关闭: 存储阵列 (512M + 32M) Byte; 页大小 (4K + 256) Byte; 块大小 64 页, 即 (256K + 16K) Byte。

- 内部 ECC 开启: 存储阵列 (512M + 16M) Byte; 页大小 (4K + 128) Byte; 块大小 64 页, 即 (256K + 8K) Byte。

串行接口: Standard SPI (CLK、CS#、DI、DO、WP#、HOLD#); Dual SPI (CLK、CS#、SIO0-SIO1、WP#、HOLD#); Quad SPI (CLK、CS#、SIO0-SIO3)。

高性能: Standard/Dual/Quad SPI 最高 166MHz; 页编程时间典型 350us; 块擦除时间典型 2ms。

低功耗: 待机电流典型 10uA; Page Read 典型 10mA; Program/Erase 典型 15mA。

高级特性: 片上 8-bit ECC; 软/硬件写保护; Unique ID; 1 个 4KB 参数页; 62 个 4KB OTP 页; 保证 Block0 为良块。

可靠性: 耐久性典型 60K 次; 数据保持 10 年 (耐久性和保持时间基于 8bit/544Byte ECC 条件)。

封装: 8-WSON (8x6mm)。

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1 General Description

The F35SQB004G is a 4G-bit (512Mx8bit) Serial NAND Flash Memory, operates on a single 3.3V VCC. The device supports the standard Serial Peripheral Interface (SPI), Dual/Quad SPI: Serial Clock, Chip Select, Serial Data SIO0 (DI), SIO1 (DO), SIO2 (WP#) and SIO3 (HOLD#). SPI clock frequencies of up to 166 MHz are supported.

The F35SQB004G supports JEDEC standard manufacturer and device ID, Unique ID, one parameter page and 62 OTP pages. An internal 8-bit ECC logic is available in the chip, which is enabled by default. The internal ECC can be disabled or enabled by command.

2 Features

Voltage Supply

- VCC: 2.7V ~ 3.6V

- Page Read: 10mA (Typ.)

- Program/Erase: 15mA (Typ.)

Organization

- ✧ Internal ECC Off:

- Memory Cell Array: (512M + 32M) Byte
- Page Size: (4k + 256) Byte
- Block Size: 64 pages, (256k + 16k) Byte

- ✧ Internal ECC On:

- Memory Cell Array: (512M + 16M) Byte
- Page Size: (4k + 128) Byte
- Block Size: 64 pages, (256k + 8k) Byte

Advanced Features

- On chip 8-Bit ECC for memory array
- Software and Hardware write protect
- Unique ID
- One 4kB parameter page
- Sixty-two 4kB OTP Pages
- Promised golden block0

Serial Interface

- Standard SPI: CLK, CS#, DI, DO, WP#, HOLD#
- Dual SPI: CLK, CS#, SIO0-SIO1, WP#, HOLD#
- Quad SPI: CLK, CS#, SIO0-SIO3

High Reliability

- Endurance: typical 60k cycles ⁽¹⁾
- Data Retention: 10 years ⁽¹⁾

Package

- 8-WSON (8x6mm)

High Performance

- 166 MHz Standard/Dual/Quad SPI clocks
- Page Program Time: 350μs (Typ.)
- Block Erase Time: 2ms (Typ.)

Note:

- (1) Endurance and Data Retention specification is based on 8bit / 544Byte ECC

Low Power

- Standby: 10μA (Typ.)

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【3 产品列表】
料号 | 容量 | I/O | 电压 | 封装 | 温度范围 | 包装
F35SQB004G-WWT | 4Gb | x1/x2/x4 | 2.7~3.6V | 8-WSON 8x6mm | -40~85 ° C | Tray
F35SQB004G-WAT | 4Gb | x1/x2/x4 | 2.7~3.6V | 8-WSON 8x6mm | -40~105 ° C | Tray

【料号编码 F35SQB004G-WWT】
F：FORESEE 品牌；35：Serial SLC NAND；S：2.7~3.6V；Q：x1/x2/x4 且支持 ECC Free；B：B 版本；004G：4Gbit；W：8-WSON；温度码 W=-40~85 ° C、A=-40~105 ° C、3=AEC-Q100 Grade 3、2=AEC-Q100 Grade 2；T：Tray 包装。

原页图示参考

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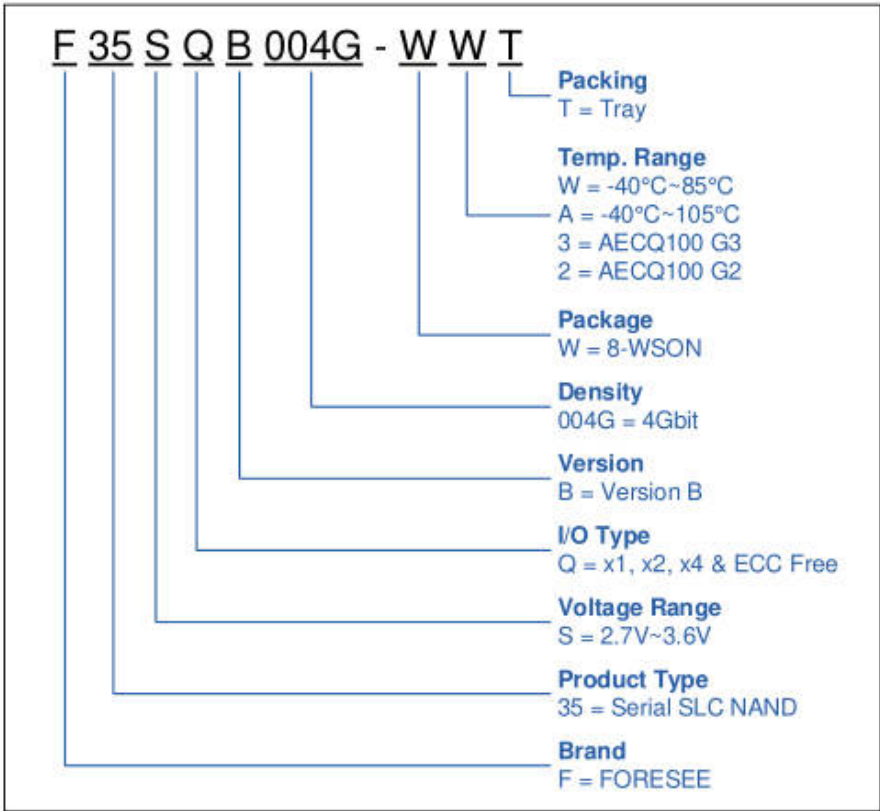


3 Product List

Table 1 Product List

Part Number	Density	I/O Type	Voltage Range	Package	Temp. Range	Packing
F35SQB004G-WWT	4Gb	x1, x2, x4	2.7V ~ 3.6V	8-WSON (8x6mm)	-40°C ~ 85°C	Tray
F35SQB004G-WAT	4Gb	x1, x2, x4	2.7V ~ 3.6V	8-WSON (8x6mm)	-40°C ~ 105°C	Tray

Figure 1 Marketing Part Numbering Chart



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【4 封装类型与引脚配置】

封装：8-WSON（8x6mm），8 引脚。

- 1 CS#
- 2 DO/SIO1
- 3 WP#/SIO2
- 4 GND
- 5 DI/SIO0
- 6 CLK
- 7 HOLD#/SIO3
- 8 VCC

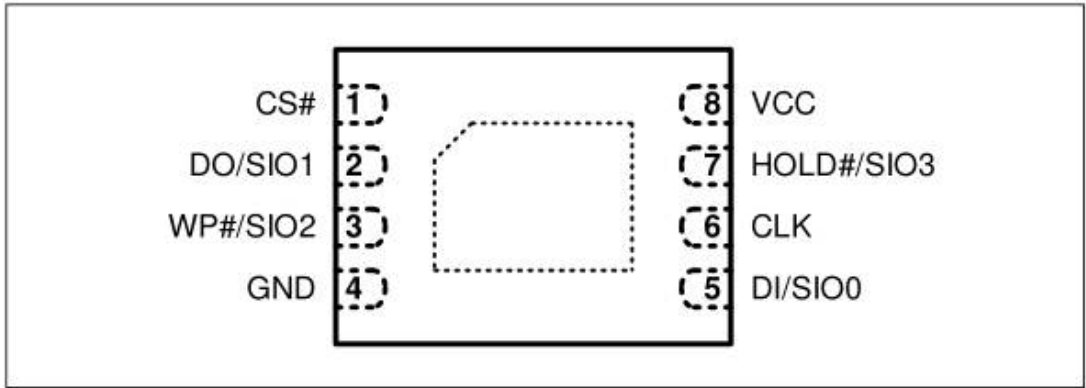
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4 Package Types and Pin Configurations

Figure 2 Pin Configuration 8-WSON (8x6mm)



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【5 引脚说明】

CS# - 片选：低电平选中器件，高电平取消选择。未选中时 DO/SIO0-SIO3 为高阻态，器件通常进入待机功耗；若内部正在执行擦除、编程或状态寄存器写周期，则内部操作继续。上电后，必须先出现一次 CS# 从高到低的跳变，器件才接受新指令。

DI、DO、SIO0-SIO3 - 串行数据输入/输出：支持标准 SPI、Dual SPI、Quad SPI。标准 SPI 使用单向 DI 在 CLK 上升沿输入命令/地址/数据，并用单向 DO 在 CLK 下降沿输出数据/状态；Dual/Quad SPI 将相应引脚配置为双向 I/O，在 CLK 上升沿输入、下降沿输出。

WP# - 写保护：可阻止状态寄存器被写入。配合 BP3-0、TB、BPRWD、SP 等保护位，可对最小 256KB（1 块）到整个阵列实施硬件写保护。

HOLD# - 保持：在 Standard/Dual SPI 且 CS# 为低时，可暂停当前操作。HOLD# 拉低后 DO 为高阻，DI/CLK 被忽略；

HOLD# 恢复高电平后继续。Quad SPI 命令执行期间，HOLD# 作为 SIO3 数据引脚，不提供 HOLD 功能。

CLK - 串行时钟输入：提供串行输入/输出的时序基准。

GND - 地。

VCC - 电源。

NC - 不连接。

原页图示参考

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5 Pin Descriptions

Table 2 Pin Description

Pin Name	Pin Functions
CS#	Chip Select The SPI Chip Select pin enables and disables device operation. When CS# is high the device is deselected and the Serial Data Output (DO or SIO0-3) pins are at high impedance. When deselected, the devices power consumption will be at standby levels unless an internal erase, program or write status register cycle is in progress. When CS# is brought low the device will be selected, power consumption will increase to active levels and instructions can be written to and data read from the device. After power-up, CS# must transition from high to low before a new instruction will be accepted.
DI, DO and SIO0-SIO3	Serial Data Input, Output and IOs The device supports standard SPI, Dual SPI and Quad SPI operation. Standard SPI instructions use the unidirectional DI (input) pin to serially write instructions, addresses or data to the device on the rising edge of the Serial Clock (CLK) input pin. Standard SPI also uses the unidirectional DO (output) to read data or status from the device on the falling edge of CLK. Dual and Quad SPI instructions use the bidirectional IO pins to serially write instructions, addresses or data to the device on the rising edge of CLK and read data or status from the device on the falling edge of CLK.
WP#	Write Protect The WP# pin can be used to prevent the Status Register from being written. Used in conjunction with the Status Register's Block Protect bits (BP3-0, TB) and Status Register Protect bits (BPRWD, SP), a portion as small as 256k-Byte (1 block) or up to the entire memory array can be hardware protected.
HOLD#	Hold During Standard and Dual SPI operations, the HOLD# pin allows the device to be paused while it is actively selected. When HOLD# is brought low, while CS# is low, the DO pin will be at high impedance and signals on the DI and CLK pins will be ignored (don't care). When HOLD# is brought high, device operation can resume. The HOLD function can be useful when multiple devices are sharing the same SPI signals. The HOLD# pin is active low. When a Quad SPI Read/Program Data Load command is issued, HOLD# pin will become a data I/O pin for the Quad operations and no HOLD function is available until the current Quad operation finishes.
CLK	Serial Clock The SPI Serial Clock Input (CLK) pin provides the timing for serial input and output operations.
GND	Ground
VCC	VCC Power Supply
NC	No Connection

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【6 框图】

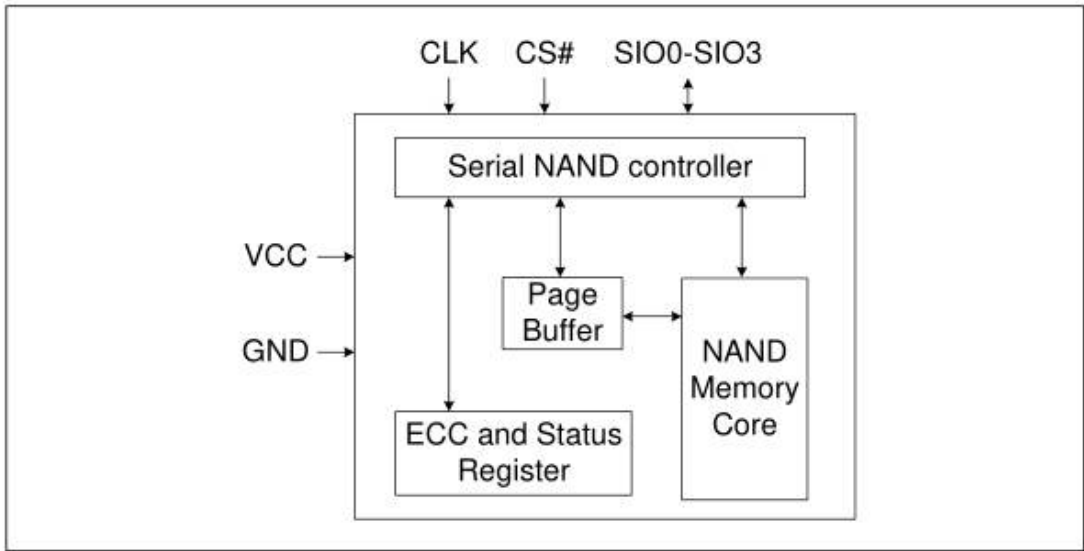
器件由串行 NAND 控制器、Page Buffer、ECC 与状态寄存器、NAND Memory Core 构成。
外部接口：CLK、CS#、SIO0-SIO3；供电：VCC/GND。
Page Buffer 位于串行接口和 NAND 核心之间，内部 ECC/状态逻辑参与读写、校验和状态管理。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

6 **Block Diagram**

Figure 3 Block Diagram



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【7 阵列组织与映射 - 阵列组织】

内部 ECC 关闭：

- 128K 页 = 2,048 块。
- 1 页 = 4K + 256 Byte。
- 1 块 = 64 页 = 256K + 16K Byte。
- 全器件 = 2,048 x (256K+16K) Byte = 4,352Mbit。

内部 ECC 开启：

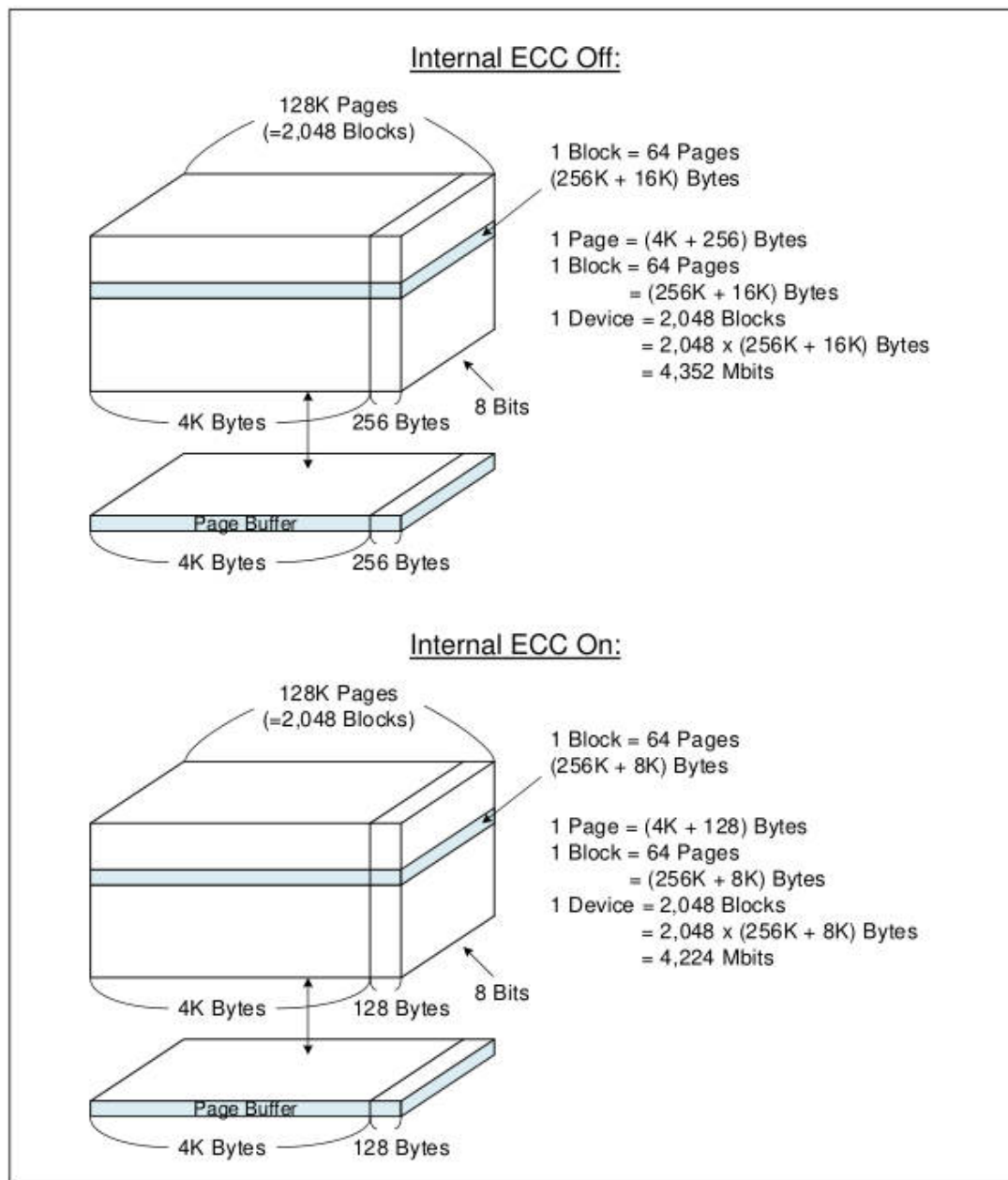
- 128K 页 = 2,048 块。
 - 用户可编程页空间 = 4K + 128 Byte（完整 spare 仍可读取）。
 - 1 块 = 64 页 = 256K + 8K Byte。
 - 全器件用户可用组织 = 2,048 x (256K+8K) Byte = 4,224Mbit。
- 数据宽度 8bit；Page Buffer 与页组织对应。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

7 Array Organization and Mapping

Figure 4 Array Organization



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【7 阵列组织与映射 - 地址映射】

块地址：PA[16:6]；页地址：PA[5:0]；字节地址：CA[12:0]。

块范围 0~2047，每块页号 0~63，有效字节地址 0~4351。

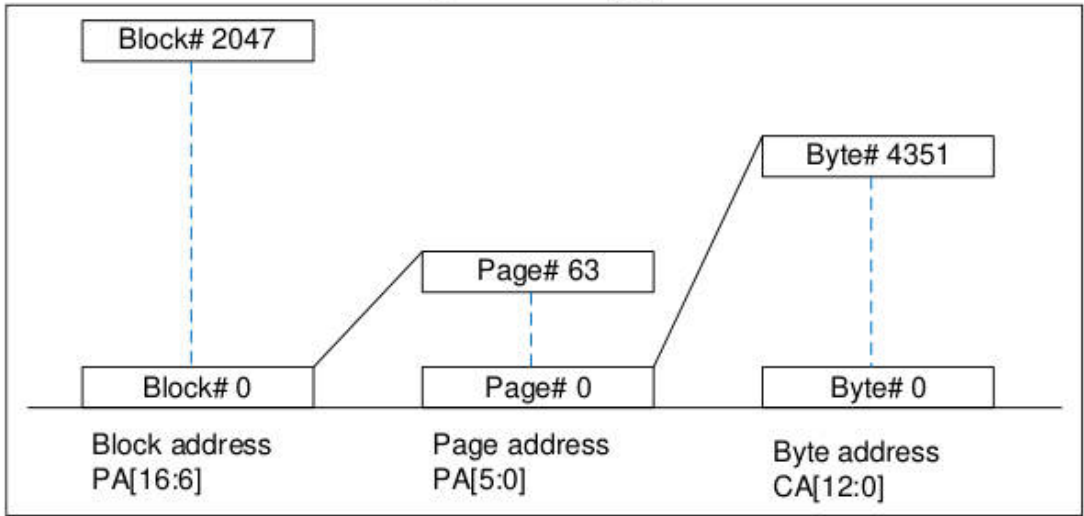
说明：

1. 13-bit 字节地址理论可表示 0~8191，但只有 0~4351 有效，其余超出边界，不能寻址。
2. 内部 ECC 关闭时，用户可读写全部 256 Byte spare 区。
3. 内部 ECC 开启时，用户可读取全部 256 Byte spare 区，但仅可编程其中前 128 Byte。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Figure 5 Address Mapping



Note:

1. The 13-bit byte address is capable of addressing from 0 to 8191 bytes. However, only bytes 0 through 4351 are valid, the rests are "out of bounds" and cannot be addressed.
2. When Internal ECC is disabled, user can read and program the entire 256 bytes spare area.
3. When Internal ECC is enabled, user can read the entire 256 bytes spare area, but can only program the first 128 bytes of the entire 256 bytes spare area.

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【8 器件操作】

8.1 一般说明：

1. 发送命令前，应通过 Get Feature 读取状态寄存器，确认器件已就绪并适合执行目标操作。
2. 输入错误命令时，器件进入待机模式并保持到下一次 CS# 下降沿；此时 SIO 为 High-Z。
3. 输入正确命令后，器件进入工作模式并保持到下一次 CS# 上升沿。

8.2 SPI 模式：支持 Mode 0（CPOL=0，CPHA=0）与 Mode 3（CPOL=1，CPHA=1）。两种模式均在 CLK 上升沿锁存输入数据，在下降沿移出数据。本文时序图均按 Mode 0 绘制。

Standard SPI：四线信号 CLK、CS#、DI、DO。

Dual SPI：使用 x2/dual IO 命令，DI/DO 变为双向 SIO0/SIO1，数据速率可达到标准 SPI 的 2 倍。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

8 Device Operation

8.1 General

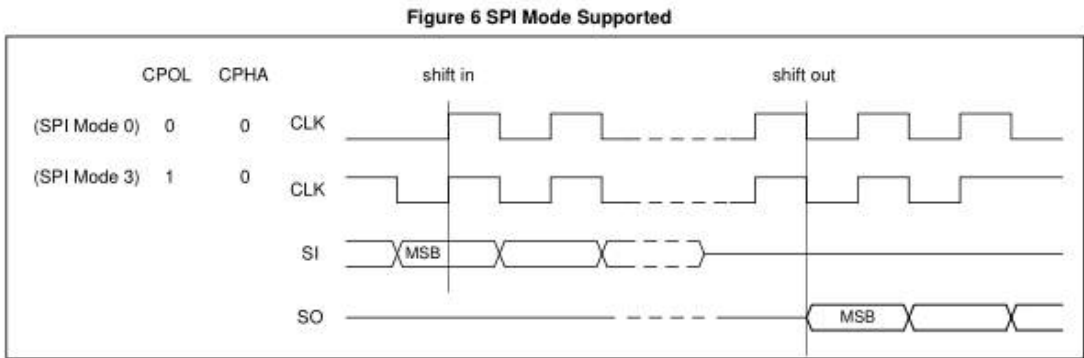
- 1. Before a command is issued, status register should be checked via get feature operations to ensure device is ready for the intended operation.
- 2. When incorrect command is inputted to this device, this device becomes standby mode and keeps the standby mode until next CS# falling edge. In standby mode, SIO pin of this device should be High-Z.
- 3. When correct command is inputted to this device, this device becomes active mode and keeps the active mode until next CS# rising edge.

8.2 SPI Modes

SPI NAND supports two SPI modes:

- CPOL = 0, CPHA = 0 (Mode 0)
- CPOL = 1, CPHA = 1 (Mode 3)

Input data is latched on the rising edge of CLK and data shifts out on the falling edge of CLK for both modes. All timing diagrams shown in this data sheet are mode 0. The difference of Mode 0 and Mode 3 is shown as **Figure 6**.



Standard SPI

SPI NAND Flash features a standard serial peripheral interface on 4 signals bus: Serial Clock (CLK), Chip Select (CS#), Serial Data Input (DI) and Serial Data Output (DO).

Dual SPI

SPI NAND Flash supports Dual SPI operation when using the x2 and dual IO commands. These commands allow data to be transferred to or from the device at two times the rate of the standard SPI. When using the Dual SPI command the DI and DO pins become bidirectional I/O pins: SIO0 and SIO1.

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【8.2 Quad SPI】

使用 x4/Quad IO 命令时支持 Quad SPI，数据吞吐可达标准 SPI 的 4 倍。此时 DI/DO 分别作为 SIO0/SIO1，WP#/HOLD# 分别作为 SIO2/SIO3。

【8.3 HOLD 功能】

Standard/Dual SPI 下，在 CS# 为低且器件已选中时，HOLD# 可暂停操作，适合多个器件共享 SPI 总线的场景。进入 HOLD 时若 CLK 已为低，则 HOLD# 下降沿立即生效；否则在下一次 CLK 下降沿生效。退出 HOLD 同理，要求 HOLD# 上升并在 CLK 低电平条件下完成。HOLD 期间 DO 高阻，DI 和 CLK 被忽略。整个 HOLD 期间 CS# 应保持低电平，以防内部逻辑状态被复位。Quad SPI 下 HOLD# 作为 SIO3，不具备 HOLD 功能。

【8.4 写保护】

器件提供多种防止误写的机制：VCC 低于阈值时复位；Write Enable/Disable 指令；擦除/编程完成后自动清除写使能。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Quad SPI

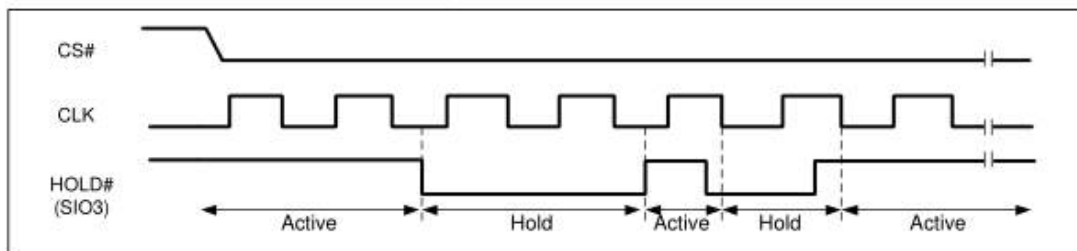
SPI NAND Flash supports Quad SPI operation when using the x4 and Quad IO commands. These commands allow data to be transferred to or from the device at four times the rate of the standard SPI. When using the Quad SPI command the DI and DO pins become bidirectional I/O pins: SIO0 and SIO1, and WP# and HOLD# pins become SIO2 and SIO3.

8.3 Hold Function

For Standard SPI and Dual SPI operations, the HOLD# signal allows the device operation to be paused while it is actively selected (when CS# is low). The Hold function may be useful in cases where the SPI data and clock signals are shared with other devices. For example, consider if the page buffer was only partially written when a priority interrupt requires use of the SPI bus. In this case the Hold function can save the state of the instruction and the data in the buffer so programming can resume where it left off once the bus is available again. The Hold function is only available for standard SPI and Dual SPI operation, not during Quad SPI. When a Quad SPI command is issued, HOLD# pin will act as a dedicated IO pin (SIO3).

To initiate a HOLD condition, the device must be selected with CS# low. A HOLD condition will activate on the falling edge of the HOLD# signal if the CLK signal is already low. If the CLK is not already low the HOLD condition will activate after the next falling edge of CLK. The HOLD condition will terminate on the rising edge of the HOLD# signal if the CLK signal is already low. If the CLK is not already low the HOLD condition will terminate after the next falling edge of CLK. During a HOLD condition, the Serial Data Output (DO) is high impedance, and Serial Data Input (DI) and serial Clock (CLK) are ignored. The Chip Select (CS#) signal should be kept active (low) for the full duration of the HOLD operation to avoid resetting the internal logic state of the device. See **Figure 7** for more details.

Figure 7 Hold Condition



8.4 Write Protection

The device provides several means to protect the data from inadvertent writes.

- Device resets when VCC is below threshold
- Write enable/disable instructions and automatic write disable after erase or program

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【8.4 写保护（续）】

- 通过 Protection Register 配合软件和 WP# 引脚实现软/硬件写保护。
- 可将 Protection Register 锁定到下一次重新上电。
- 可使用 Protection Register 对存储阵列实施 OTP 类型写保护。

上电后 WEL=0，器件处于写禁止状态。执行 Program Execute 或 Block Erase 前必须先发送 Write Enable。每次编程或擦除完成后，WEL 自动清零。

软件写保护由 Write Status/Set Feature 类操作配合 BPRWD、SP、TB、BP3-0 等位控制，可将部分或全部存储阵列配置为只读。结合 WP# 引脚还可硬件禁止/允许对 Protection Register 的修改。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

- Software and Hardware (WP# pin) write protection using Protection Register
- Lock Down write protection for Protection Register until the next power-up
- One Time Program (OTP) write protection for memory array using Protection Register

After power-up the device is automatically placed in a write-disabled state with the Status Register Write Enable Latch (WEL) set to a 0. A Write Enable instruction must be issued before a Program Execute or Block Erase instruction will be accepted. After completing a program or erase instruction the Write Enable Latch (WEL) is automatically cleared to a write-disabled state of 0.

Software controlled write protection is facilitated using the Write Status Register instruction and setting the Status Register Protect (BPRWD, SP) and Block Protect (TB, BP3-0) bits. These settings allow a portion or the entire memory array to be configured as read only. Used in conjunction with the WP# pin, changes to the Status Register can be enabled or disabled under hardware control. See **Protection Register** for further information.

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【9 状态寄存器】

Protection、Configuration、Status 寄存器均通过 Get Feature (0Fh) /Set Feature (1Fh) + 1 Byte 寄存器地址访问。Sector ECC Status 只能通过 Get Feature (0Fh) 读取。

寄存器位：

A0h Protection：Bit7 BPRWD；Bit6 BP3；Bit5 BP2；Bit4 BP1；Bit3 BP0；Bit2 TB；Bit1 R；Bit0 SP。

B0h Configuration：Bit7 OTP-L；Bit6 OTP-E；Bit5 R；Bit4 ECC-E；Bit3 R；Bit2 DRV1；Bit1 DRV0；Bit0 QE。

C0h Status：Bit7 R；Bit6 ECCS2；Bit5 ECCS1；Bit4 ECCS0；Bit3 P-FAIL；Bit2 E-FAIL；Bit1 WEL；Bit0 OIP。

D0h Configuration：Bit0 ECC-M，其余保留。

10h~80h：Sector0~Sector7 ECC Status，Bit7~4 为 sector 标识，Bit3~0 为 SxES3~0。

R 为保留位，无功能，读值可能为 0 或 1，应忽略；Set Feature 时可写 0，但不会产生作用。

FFh Reset 不会清除此前的 Feature 设置；除表中明确说明者外，相关位保持到掉电重启或由软件修改。复位期间 OIP=1，完成后回到 0。Reset 不影响 Block Protection 与 Configuration 寄存器。

上电后保护位 BP3-0/TB 默认 11111（全阵列保护）。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

9 Status Registers

For Protection Register, Configuration Register and Status Register, each register is accessed by Get Feature (0Fh) and Set Feature (1Fh) commands combined with 1-Byte Register Address respectively. For Sector ECC Status Registers, each one can only be accessed by Get Feature (0Fh) command combined with 1-Byte Register Address.

Table 3 Status Registers

Register	Address	Data Bits							
		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Protection	A0h	BPRWD	BP3	BP2	BP1	BP0	TB	R	SP
Configuration	B0h	OTP-L	OTP-E	R	ECC-E	R	DRV1	DRV0	QE
Status	C0h	R	ECCS2	ECCS1	ECCS0	P-FAIL	E-FAIL	WEL	OIP
Configuration	D0h	R	R	R	R	R	R	R	ECC-M
Sector0 ECC Status	10h	0	0	0	0	S0ES3	S0ES2	S0ES1	S0ES0
Sector1 ECC Status	20h	0	0	0	1	S1ES3	S1ES2	S1ES1	S1ES0
Sector2 ECC Status	30h	0	0	1	0	S2ES3	S2ES2	S2ES1	S2ES0
Sector3 ECC Status	40h	0	0	1	1	S3ES3	S3ES2	S3ES1	S3ES0
Sector4 ECC Status	50h	0	1	0	0	S4ES3	S4ES2	S4ES1	S4ES0
Sector5 ECC Status	60h	0	1	0	1	S5ES3	S5ES2	S5ES1	S5ES0
Sector6 ECC Status	70h	0	1	1	0	S6ES3	S6ES2	S6ES1	S6ES0
Sector7 ECC Status	80h	0	1	1	1	S7ES3	S7ES2	S7ES1	S7ES0

Note:

- (1) R: Reserved Bit and has no function. They may be read out as a "0" or "1". It is recommended to ignore the values of those bits. During a Set Feature command, the Reserved Bits can be written as "0", but there will not be any effects.

The Reset command (FFh) will not clear the previous feature setting, the feature setting data bits remain until the power is being cycled or modified by the settings in the table below. After a Reset command is issued, the OIP bit will go high. This bit can be polled to determine when the Reset operation is complete, as it will return to the default value (0) after the reset operation is finished. Issuing the Reset command has no effect on the Block Protection and Configuration registers.

Table 4 Default Values of the Status Registers after power up and Device Reset

Register	Address	Bits	Shipment Default	Power Up	After Reset Command
Protection	A0h	BP3-0, TB	1 1 1 1 1	1 1 1 1 1	No Change

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【状态寄存器默认值与 9.1 保护寄存器】

上电/复位要点：

- BPRWD=0、SP=0；OTP-L 若已永久锁定则为 1，否则 0；OTP-E=0；ECC-E=1；QE=0；DRV1-0=00。
- C0h：P-FAIL=0、E-FAIL=0、WEL=0、OIP=0；上电后 ECCS2-0 反映 Block0/Page0 的 ECC 状态，Reset 后清 000。
- D0h：ECC-M 上电默认 1，Reset 不改变。
- Sector0~7 ECC 状态上电反映 Block0/Page0 对应 sector 状态，Reset 后清 0000。

9.1.1 Block Protect (BP3-0、TB)：易失可读写，用于控制受 Program/Erase 保护的区域。可通过 Set Feature 修改。可保护全阵列、部分阵列或不保护。上电默认全 1，保护整个阵列。

9.1.2 Status Register Protect (BPRWD、SP)：易失可读写，用于选择软件保护、硬件保护和本次供电周期锁定等保护方式。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Register	Address	Bits	Shipment Default	Power Up	After Reset Command
Configuration	B0h	BPRWD	0	0	No Change
		SP	0	0	No Change
		OTP-L	0	Loked:1, else 0	No Change
		OTP-E	0	0	No Change
		ECC-E	1	1	No Change
		QE	0	0	No Change
		DRV1-0	0 0	0 0	No Change
Status	C0h	ECCS2-0	0 0 0	Status of Page0/Block0	0 0 0
		P-FAIL	0	0	0
		E-FAIL	0	0	0
		WEL	0	0	0
		OIP	0	0	0
Configuration	D0h	ECC-M	1	1	No Change
Sector0 ECC Status	10h	S0ES3-0	0 0 0 0	Sector0 ECC Status of Page0/Block0	0 0 0 0
Sector1 ECC Status	20h	S1ES3-0	0 0 0 0	Sector1 ECC Status of Page0/Block0	0 0 0 0
Sector2 ECC Status	30h	S2ES3-0	0 0 0 0	Sector2 ECC Status of Page0/Block0	0 0 0 0
Sector3 ECC Status	40h	S3ES3-0	0 0 0 0	Sector3 ECC Status of Page0/Block0	0 0 0 0
Sector4 ECC Status	50h	S4ES3-0	0 0 0 0	Sector4 ECC Status of Page0/Block0	0 0 0 0
Sector5 ECC Status	60h	S5ES3-0	0 0 0 0	Sector5 ECC Status of Page0/Block0	0 0 0 0
Sector6 ECC Status	70h	S6ES3-0	0 0 0 0	Sector6 ECC Status of Page0/Block0	0 0 0 0
Sector7 ECC Status	80h	S7ES3-0	0 0 0 0	Sector7 ECC Status of Page0/Block0	0 0 0 0

9.1 Protection Register

9.1.1 Block Protect Bits (BP3-0, TB)

The Block Protect bits (BP3-0, TB) are volatile read/write bits that provide Write Protection control and status. Block Protect bits can be set using the Set Feature Instruction. All, none or a portion of the memory array can be protected from Program and Erase instructions (See **Table 6**). The default values for the Block Protection bits are 1 after power up to protect the entire array.

9.1.2 Status Register Protect Bits (BPRWD, SP)

The Status Register Protect bits (BPRWD, SP) are volatile read/write bits which control the method of

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【9.1.2 状态寄存器保护】

BPRWD/SP/QE/WP# 组合：

- SP=0、QE=1：Protection Register 可改；WP# 不再有保护功能，WP# 引脚恒作为 SIO2。
- SP=1：当前供电周期内 Protection Register 不可改；掉电再上电后（BPRWD,SP）回到（0,0）。
- BPRWD=0、SP=0、QE=0：Protection Register 可改。
- BPRWD=1、SP=0、QE=0、WP#=0：不可改。
- BPRWD=1、SP=0、QE=0、WP#=1：可改。

【9.1.3 存储区块保护】

TB/BP3-0 选择从高地址端（TB=0）或低地址端（TB=1）保护。可选保护比例从 1/2048（256KB）逐级扩大到 1/2（256MB），BP3=BP2=1 时保护全部 512MB。BP3-0=0000 时不保护。

上端保护示例：256KB 对应 Block2047/PA 1FFC0h~1FFFFh；512KB 对应 Block2046~2047；1MB 对应 Block2044~2047；依次扩大至上半区 256MB。

下端保护示例：256KB 对应 Block0/PA 00000h~0003Fh；512KB 对应 Block0~1；1MB 对应 Block0~3；依次扩大至下半区 256MB。

若 Erase/Program 命令指定的区域包含任何受保护数据，该命令将被忽略。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

write protection: software protection, hardware protection, power supply lock-down.

Table 5 Status Register Protection

BPRWD	SP	QE	WP#	Descriptions
X	0	1	X	Protection Register can be changed No WP# functionality, WP# pin will always function as SIO2
X	1	X	X	Protection Register cannot be changed during the current power cycle
0	0	0	X	Protection Register can be changed
1	0	0	0	Protection Register can NOT be changed
1	0	0	1	Protection Register can be changed

Note:

(1) When SP =1, a power-down, power-up cycle will change (BPRWD, SP) to (0, 0) state.

9.1.3 Status Register Memory Protection

Table 6 Block Protection Bits

TB	BP3	BP2	BP1	BP0	PROTECTED BLOCK(S)	PROTECTED PAGE ADDRESS PA[16:0]	PROTECTED DENSITY	PROTECTED PORTION
X	0	0	0	0	NONE	NONE	NONE	NONE
0	0	0	0	1	2047	1FFC0h – 1FFFFh	256KB	Upper 1/2048
0	0	0	1	0	2046 & 2047	1FF80h – 1FFFFh	512KB	Upper 1/1024
0	0	0	1	1	2044 thru 2047	1FF00h – 1FFFFh	1MB	Upper 1/512
0	0	1	0	0	2040 thru 2047	1FE00h – 1FFFFh	2MB	Upper 1/256
0	0	1	0	1	2032 thru 2047	1FC00h – 1FFFFh	4MB	Upper 1/128
0	0	1	1	0	2016 thru 2047	1F800h – 1FFFFh	8MB	Upper 1/64
0	0	1	1	1	1984 thru 2047	1F000h – 1FFFFh	16MB	Upper 1/32
0	1	0	0	0	1920 thru 2047	1E000h – 1FFFFh	32MB	Upper 1/16
0	1	0	0	1	1792 thru 2047	1C000h – 1FFFFh	64MB	Upper 1/8
0	1	0	1	0	1536 thru 2047	18000h – 1FFFFh	128MB	Upper 1/4
0	1	0	1	1	1024 thru 2047	10000h – 1FFFFh	256MB	Upper 1/2
1	0	0	0	1	0	00000h – 0003Fh	256KB	Lower 1/2048
1	0	0	1	0	0 & 1	00000h – 0007Fh	512KB	Lower 1/1024
1	0	0	1	1	0 thru 3	00000h – 000FFh	1MB	Lower 1/512
1	0	1	0	0	0 thru 7	00000h – 001FFh	2MB	Lower 1/256
1	0	1	0	1	0 thru 15	00000h – 003FFh	4MB	Lower 1/128
1	0	1	1	0	0 thru 31	00000h – 007FFh	8MB	Lower 1/64
1	0	1	1	1	0 thru 63	00000h – 00FFFh	16MB	Lower 1/32
1	1	0	0	0	0 thru 127	00000h – 01FFFh	32MB	Lower 1/16
1	1	0	0	1	0 thru 255	00000h – 03FFFh	64MB	Lower 1/8
1	1	0	1	0	0 thru 511	00000h – 07FFFh	128MB	Lower 1/4
1	1	0	1	1	0 thru 1023	00000h – 0FFFFh	256MB	Lower 1/2
X	1	1	X	X	0 thru 2047	00000h – 1FFFFh	512MB	ALL

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【9.2 配置寄存器】

9.2.1 OTP-L：非易失锁定位。器件提供 62 个完整 OTP 页，出厂为 FFh。OTP 只能把位从 1 编程为 0，不能擦除。写入并验证关键数据后，可将 OTP-L 置 1，永久锁定整个 OTP 区，防止后续修改。

9.2.2 OTP-E：访问 OTP/UID/Parameter 区前必须置 1。置 1 后可用标准 Program/Read 命令访问 OTP，并读取 Unique ID 和 Parameter Page。上电或 RESET 后默认 0。

9.2.3 ECC-E：内部 ECC 使能。页编程时计算 ECC 并存入每页额外 128 Byte 区；读取时根据已存 ECC 校验并进行必要纠错，结果由 ECC Status 位指示。上电默认 ECC-E=1，Device Reset 不改变。

9.2.4 DRV1-0 输出驱动强度：00=100%（默认）；01=75%；10=50%；11=25%。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Note:

- (1) X = don't care
- (2) If any Erase or Program command specifies a memory region that contains protected data portion, this command will be ignored.

9.2 Configuration Register

9.2.1 One Time Program Lock Bit (OTP-L)

OTP-L is non-volatile.

The device provides an OTP area for the system to store critical data that cannot be changed once it's locked. The OTP area consists of 62 full pages. The default data in the OTP area are FFh. Only Program command can be issued to the OTP area to change the data from "1" to "0", and the OTP area cannot be erased.

Once the correct data is programmed in and verified, the system developer can set OTP-L bit to 1, so that the entire OTP area will be locked to prevent further alteration to the data.

9.2.2 Enter OTP Access Mode Bit (OTP-E)

The OTP-E bit must be set to 1 in order to use the standard Program/Read commands to access the OTP area as well as to read the Unique ID / Parameter Page information. The default value after power up or a RESET command is 0.

9.2.3 ECC Enable Bit (ECC-E)

The device has a built-in ECC algorithm that can be used to preserve the data integrity. Internal ECC calculation is done during page programming, and the result is stored in the extra 128-Byte area for each page. During the data read operation, ECC engine will verify the data values according to the previously stored ECC information and to make necessary corrections if needed. The verification and correction status is indicated by the ECC Status Bits. ECC function is enabled by default when power on (ECC-E=1), and it will not be changed by the Device Reset command.

9.2.4 Output Driver Strength (DRV1-0)

Table 7 Output Driver Strength

DRV1	DRV0	Output Driver Strength
0	0	100% (default)
0	1	75%
1	0	50%
1	1	25%

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【9.2.5 Quad 使能 QE】

QE 为易失位。QE=0（出厂默认）时使用非 Quad 模式，WP#/HOLD# 功能有效；QE=1 时启用 Quad I/O，WP#/HOLD# 功能被禁用并作为数据 I/O。重新上电后 QE 回到 0。

【9.2.6 ECC 模式 ECC-M】

ECC-M 为易失位，用于选择两种内部 ECC 保护范围：

- ECC-M=0：整个 sector 均受 ECC 保护。
- ECC-M=1：每个 sector 的 spare 区前 4 Byte 不受 ECC 保护，其余部分受保护。

重新上电后 ECC-M 默认 1；FFh Reset 不改变该位。

【9.3.1 ECC 状态 ECCS2-0】

ECCS2-0 在读操作结束后反映页面内各 sector 的纠错状态；ECC-E=0 时其值无意义。RESET 后清 0。上电复位后该值反映 Block0/Page0 的 ECC 状态。

000：未检测到位错误；001：≤3bit 错误且已纠正；010：4bit 已纠正；011：5bit 已纠正；100：6bit 已纠正；101：7bit 已纠正；110：8bit 已纠正；111：>8bit，至少一个 sector 无法纠正。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

9.2.5 Quad Enable Bit (QE)

The Quad Enable (QE) bit is a volatile bit, while it is "0" (factory default), it performs non-Quad and WP#, HOLD# are enabled. While QE is "1", it performs Quad I/O mode and WP#, HOLD# are disabled. In another word, if the system goes into four I/O mode (QE=1), the WP# and HOLD# function will be disabled. Upon power cycle, the QE bit will go into the factory default setting "0".

9.2.6 ECC Mode Bit (ECC-M)

The volatile ECC-M bit is used to switch internal ECC protection modes. According to requirement, customer can select one of two ECC modes which are described as below.

When ECC-M=0, the entire sector is protected by ECC.

When ECC-M=1, the first four bytes of spare area in each sector are NOT protected by ECC while the rests are protected.

Please refer to **Sector ECC Status Register** for detailed information about ECC sector.

After a power cycle, the ECC-M bit will be set to the factory default value "1". Reset command (FFh) has no effect on ECC-M bit.

9.3 Status Register

9.3.1 ECC Status Bit (ECCS2-0)

ECC function is used in NAND flash memory to correct limited memory errors during read operations. The ECC Status Bits (ECCS2, ECCS1, ECCS0) should be checked after the completion of a Read operation to verify the data integrity. The ECC Status bits values are don't care if ECC-E=0. These bits will be cleared to 0 after a RESET command.

The ECCS2-0 value reflects the ECC status of the content of the page 0 of block 0 after a power-on reset.

Table 8 ECC Bits Status

ECCS2	ECCS1	ECCS0	Description
0	0	0	No bit errors were detected during the previous read operation
0	0	1	Bit errors (<=3) were detected in one or more sectors and were corrected
0	1	0	Bit errors (=4) were detected in one or more sectors and were corrected
0	1	1	Bit errors (=5) were detected in one or more sectors and were corrected
1	0	0	Bit errors (=6) were detected in one or more sectors and were corrected
1	0	1	Bit errors (=7) were detected in one or more sectors and were corrected
1	1	0	Bit errors (=8) were detected in one or more sectors and were corrected
1	1	1	Bit errors (>8) were detected in one or more sectors and cannot be corrected

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【9.3.2 Program/Erase Failure】

P-FAIL/E-FAIL 指示内部 Program/Erase 是否成功。对受保护块或已锁 OTP 区执行 Program 时也会置 P-FAIL；对受保护块执行 Erase 时也会置 E-FAIL。开始新的 Program Execute 或 Block Erase，以及 Device Reset 时清除相应失败位。

【9.3.3 WEL】

WEL 为只读状态位。执行 Write Enable 后置 1；上电或执行 Write Disable、Program Execute、Block Erase、Page Data Read，以及 OTP 页 Program Execute 后回到 0。

【9.3.4 OIP】

OIP 为只读位。上电、Page Read、Program Execute、Block Erase、OTP Locking 期间为 1。此时除 Get Feature 和 Soft Reset 外，其他指令被忽略。操作完成后 OIP=0，表示可接收下一条指令。

【9.4 Sector ECC 状态寄存器】

每个 ECC sector = 512 Byte main + 32 Byte spare；每页共 8 个 sector。Sector ECC Status Register 给出本次读取中每个 sector 的错误数量级。

4KB 主区被分为 Main0~Main7，每区 512B；spare 区按 Spare0~Spare7 对应各 sector，包含用户 Meta Data 和 ECC Parity Data。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

9.3.2 Program/Erase Failure (P-FAIL, E-FAIL)

The Program/Erase Failure Bits are used to indicate whether the internally-controlled Program/Erase operation was executed successfully or not. P-FAIL bit will also be set when the Program command is issued to a protected block or locked OTP area, and E-FAIL bit will also be set when the Erase command is issued to a protected block. Both bits will be cleared at the beginning of the Program Execute or Block Erase instructions as well as the device Reset command.

9.3.3 Write Enable Latch (WEL)

Write Enable Latch (WEL) is a read only bit. The WEL bit is set to 1 after executing a Write Enable Instruction. The WEL bit is cleared to 0 when the device is write disabled. A write disable state occurs upon power-up or after any of the following instructions: Write Disable, Program Execute, Block Erase, Page Data Read, and Program Execute for OTP pages.

9.3.4 Operation in Progress (OIP)

OIP is a read only bit that is set to a 1 state when the device is powering up or executing a Page Read, Program Execute, Block Erase and OTP Locking. During this time the device will ignore further instructions except for the Get Feature or Soft Reset instructions. When the program, erase or page read instruction has completed, the OIP bit will be cleared to a 0 state indicating the device is ready for further instructions.

9.4 Sector ECC Status Register

A sector is composed by a 512 Byte main area and a 32 Byte spare area, so a page has eight sectors. The Sector ECC Status Register indicates the number of errors in each sector as identified from an ECC check during a read operation.

Table 9 4KByte Page Assignment

Main Area (4KB)									Spare Area (256B)															
User Data									User Meta Data (I + II)								ECC Parity Data							
Main 0	Main 1	Main 2	Main 3	Main 4	Main 5	Main 6	Main 7		Spare 0	Spare 1	Spare 2	Spare 3	Spare 4	Spare 5	Spare 6	Spare 7	Spare 0	Spare 1	Spare 2	Spare 3	Spare 4	Spare 5	Spare 6	Spare 7
512B	512B	512B	512B	512B	512B	512B	512B		4B+12B	4B+12B	4B+12B	4B+12B	4B+12B	4B+12B	4B+12B	4B+12B	16B	16B	16B	16B	16B	16B	16B	16B

Table 10 Area Address and ECC Protection

Byte Address Range	Area	ECC Protection		Description
		ECC-M = 0	ECC-M = 1	
0000h ~ 01FFh	Main 0	Yes	Yes	User Data 0

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【表10 区域地址与 ECC 保护】
Main0~Main7：地址 0000h~0FFFh，每 0x200 Byte 一个 512B 主数据 sector；无论 ECC-M=0/1 均受 ECC 保护。
每个 Spare sector 的前 4 Byte（如 1000h~1003h、1010h~1013h……1070h~1073h）：ECC-M=0 时保护，ECC-M=1 时不保护，作为 User Meta Data I。
对应后 12 Byte（1004h~100Fh……1074h~107Fh）：两种 ECC-M 均保护，作为 User Meta Data II。
1080h~10FFh：每 sector 16 Byte ECC Parity Data，均由 ECC 保护/使用。
【表11 ECC Sector 定义】
Sector0 = Main0 + Spare0；Sector1 = Main1 + Spare1；……；Sector7 = Main7 + Spare7。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

0200h ~ 03FFh	Main 1	Yes	Yes	User Data 1
0400h ~ 05FFh	Main 2	Yes	Yes	User Data 2
0600h ~ 07FFh	Main 3	Yes	Yes	User Data 3
0800h ~ 09FFh	Main 4	Yes	Yes	User Data 4
0A00h ~ 0BFFh	Main 5	Yes	Yes	User Data 5
0C00h ~ 0DFFh	Main 6	Yes	Yes	User Data 6
0E00h ~ 0FFFh	Main 7	Yes	Yes	User Data 7
1000h ~ 1003h	Spare 0	Yes	No	User Meta 0 Data I
1004h ~ 100Fh	Spare 0	Yes	Yes	User Meta 0 Data II
1010h ~ 1013h	Spare 1	Yes	No	User Meta 1 Data I
1014h ~ 101Fh	Spare 1	Yes	Yes	User Meta 1 Data II
1020h ~ 1023h	Spare 2	Yes	No	User Meta 2 Data I
1024h ~ 102Fh	Spare 2	Yes	Yes	User Meta 2 Data II
1030h ~ 1033h	Spare 3	Yes	No	User Meta 3 Data I
1034h ~ 103Fh	Spare 3	Yes	Yes	User Meta 3 Data II
1040h ~ 1043h	Spare 4	Yes	No	User Meta 4 Data I
1044h ~ 104Fh	Spare 4	Yes	Yes	User Meta 4 Data II
1050h ~ 1053h	Spare 5	Yes	No	User Meta 5 Data I
1054h ~ 105Fh	Spare 5	Yes	Yes	User Meta 5 Data II
1060h ~ 1063h	Spare 6	Yes	No	User Meta 6 Data I
1064h ~ 106Fh	Spare 6	Yes	Yes	User Meta 6 Data II
1070h ~ 1073h	Spare 7	Yes	No	User Meta 7 Data I
1074h ~ 107Fh	Spare 7	Yes	Yes	User Meta 7 Data II
1080h ~ 108Fh	Spare 0	Yes	Yes	ECC Parity Data 0
1090h ~ 109Fh	Spare 1	Yes	Yes	ECC Parity Data 1
10A0h ~ 10AFh	Spare 2	Yes	Yes	ECC Parity Data 2
10B0h ~ 10BFh	Spare 3	Yes	Yes	ECC Parity Data 3
10C0h ~ 10CFh	Spare 4	Yes	Yes	ECC Parity Data 4
10D0h ~ 10DFh	Spare 5	Yes	Yes	ECC Parity Data 5
10E0h ~ 10EFh	Spare 6	Yes	Yes	ECC Parity Data 6
10F0h ~ 10FFh	Spare 7	Yes	Yes	ECC Parity Data 7

Table 11 Definition of ECC Sector

Sector	Main Area	Spare Area
Sector 0	Main 0	Spare 0
Sector 1	Main 1	Spare 1
Sector 2	Main 2	Spare 2
Sector 3	Main 3	Spare 3
Sector 4	Main 4	Spare 4
Sector 5	Main 5	Spare 5
Sector 6	Main 6	Spare 6
Sector 7	Main 7	Spare 7

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【9.4.1 Sector 信息】

Sector ECC Status Register 的 Bit7~4 表示 sector 编号：0000=Sector0, 0001=Sector1, 0010=Sector2, 0011=Sector3, 0100=Sector4, 0101=Sector5, 0110=Sector6, 0111=Sector7；其他值保留。

【9.4.2 Sector ECC 状态】

Bit3~0：

0000 无位错误；0001 1bit 已纠正；0010 2bit 已纠正；0011 3bit 已纠正；0100 4bit 已纠正；0101 5bit 已纠正；0110 6bit 已纠正；0111 7bit 已纠正；1000 8bit 已纠正；1001 >8bit, 无法纠正；其他值保留。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Table 12 Sector ECC Status Register 0-7

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Sector Information				Sector ECC Status			

9.4.1 Sector Information

Table 13 Sector Information

Bit 7 ~ Bit 4	Sector Information
0000	Sector 0
0001	Sector 1
0010	Sector 2
0011	Sector 3
0100	Sector 4
0101	Sector 5
0110	Sector 6
0111	Sector 7
Others	Reserved

9.4.2 Sector ECC Status

Table 14 Sector ECC Status

Bit 3 ~ Bit 0	Sector ECC Status
0000	No bit error was detected during the previous read operation
0001	1-bit error was detected in the sector and was corrected
0010	2-bit errors were detected in the sector and were corrected
0011	3-bit errors were detected in the sector and were corrected
0100	4-bit errors were detected in the sector and were corrected
0101	5-bit errors were detected in the sector and were corrected
0110	6-bit errors were detected in the sector and were corrected
0111	7-bit errors were detected in the sector and were corrected
1000	8-bit errors were detected in the sector and were corrected
1001	More than 8-bit errors were detected in the sector and cannot be corrected
Others	Reserved

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【10 命令 - 10.1 命令集】

Soft RESET FFh

Read JEDEC ID 9Fh + Dummy + MID + DID1 + DID2

Get Feature 0Fh + SR Addr -> S7-0...

Set Feature 1Fh + SR Addr + S7-0

Write Enable 06h

Write Disable 04h

Block Erase D8h + PA23-16 + PA15-8 + PA7-0

Program Data Load 02h + CA15-8 + CA7-0 + Data...

Random Program Data Load 84h + CA + Data...

Quad Program Data Load 32h + CA + x4 Data...

Random Quad Program Data Load 34h + CA + x4 Data...

Program Execute 10h + PA

Page Read(to cache) 13h + PA

Read From Cache 03h/0Bh + CA + Dummy + Data...

Read From Cache x2 3Bh + CA + Dummy + x2 Data...

Read From Cache x4 6Bh + CA + Dummy + x4 Data...

说明：CA 仅 CA[12:0] 有效，CA[15:13] 为 dummy；PA 仅 PA[16:0] 有效，PA[23:17] 为 dummy。PA[16:6] 为 2,048 个 256KB 块地址，PA[5:0] 为每块 64 个 4KB 页地址。

Dual 输出：SIO0=D6,D4,D2,D0...；SIO1=D7,D5,D3,D1...

Quad I/O：SIO0=D4,D0...；SIO1=D5,D1...；SIO2=D6,D2...；SIO3=D7,D3...

QE=0 时所有 Quad Program/Read 命令禁用。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

10 Commands

10.1 Command Set

Table 15 Command Set

Commands	Byte1	Byte2	Byte3	Byte4	Byte5	ByteN
Soft RESET	FFh					
Read JEDEC ID	9Fh	Dummy	MID	DID	DID	
Get Feature	0Fh	SR Addr	S7-0	S7-0	S7-0	S7-0
Set Feature	1Fh	SR Addr	S7-0			
Write Enable	06h					
Write Disable	04h					
Block Erase	D8h	PA23-16	PA15-8	PA7-0		
Program Data Load	02h	CA15-8	CA7-0	D7-D0	Next Byte	
Random Program Data Load	84h	CA15-8	CA7-0	D7-D0	Next Byte	
Quad Program Data Load	32h	CA15-8	CA7-0	D7-D0 / 4	Next Byte	...
Random Quad Program Data Load	34h	CA15-8	CA7-0	D7-D0 / 4	Next Byte	...
Program Execute	10h	PA23-16	PA15-8	PA7-0		
Page Read (to cache)	13h	PA23-16	PA15-8	PA7-0		
Read From Cache	03h or 0Bh	CA15-8	CA7-0	Dummy	D7-D0	Next Byte
Read From Cache x 2	3Bh	CA15-8	CA7-0	Dummy	D7-D0 / 2	Next Byte
Read From Cache x 4	6Bh	CA15-8	CA7-0	Dummy	D7-D0 / 4	Next Byte

Note:

- (1) Output designates data output from the device.
- (2) Column Address (CA) only requires CA [12:0], CA [15:13] are considered as dummy bits.
- (3) Page Address (PA) only requires PA [16:0], PA [23:17] are considered as dummy bits. PA [16:6] is the address for 256kB blocks (total 2,048 blocks), PA [5:0] is the address for 4kB pages (total 64 pages for each block).
- (4) Dual SPI Data Output (D7-D0 / 2) format:
SIO0 = D6, D4, D2, D0...
SIO1 = D7, D5, D3, D1...
- (5) Quad SPI Data Input / Output (D7-D0 / 4) format:
SIO0 = D4, D0
SIO1 = D5, D1
SIO2 = D6, D2
SIO3 = D7, D3
- (6) All Quad Program/Read commands are disabled when QE bit is set to 0 in the Configuration Register.

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【10.2 软复位 (FFh)】

器件接受 Reset 后约需 t_{RST} 完成复位；根据当前操作不同， t_{RST} 约 5us~200us。此期间不接受其他命令。

若接受 Reset 时内部正在执行 Erase/Program，可能造成数据损坏，因此建议发送 Reset 前先读取 Status Register 的 OIP，确认内部操作已结束。

图8给出了 FFh Soft Reset 以及随后用 Get Feature 轮询状态的时序。

原页图示参考

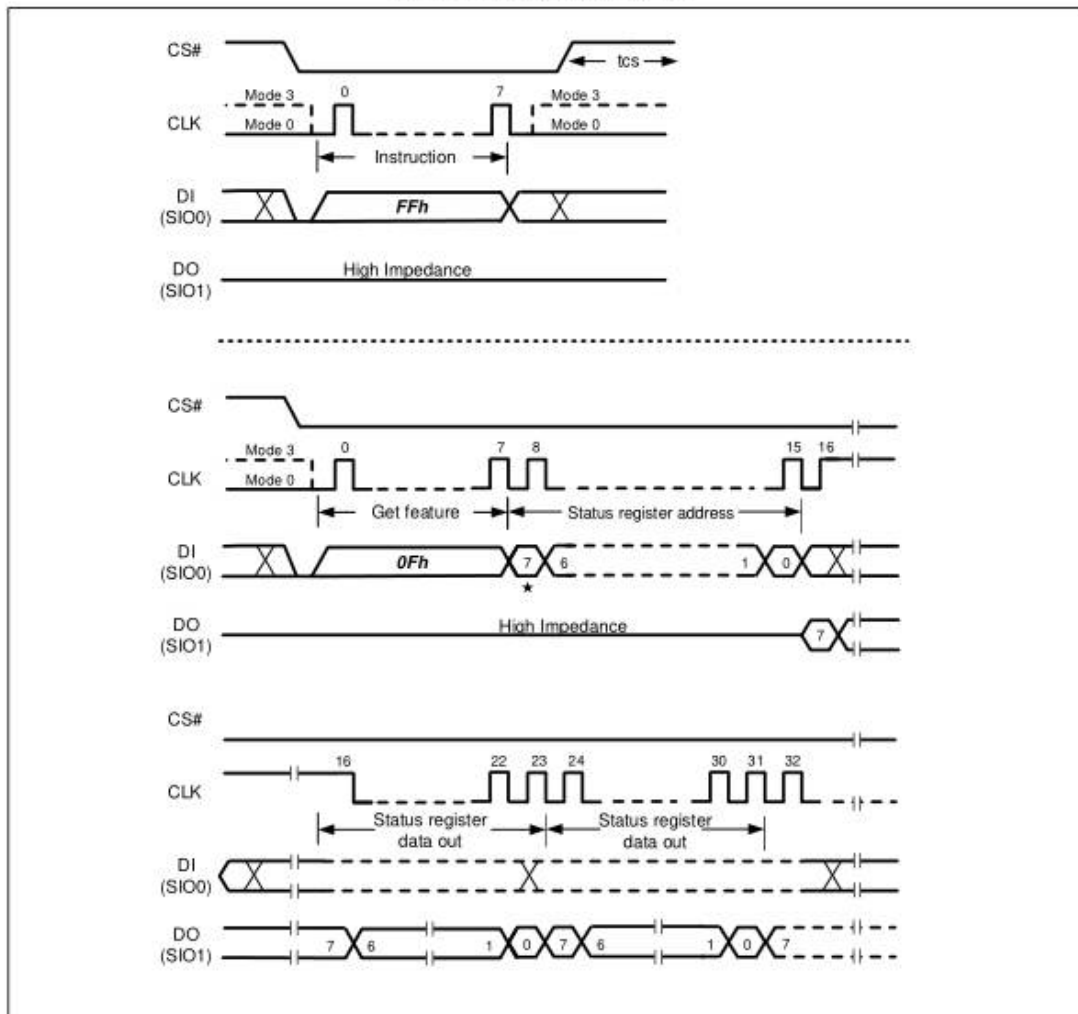
以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

10.2 Soft Reset (FFh)

Once the Reset command is accepted by the device, the device will take approximately t_{RST} to reset, depending on the current operation the device is performing, t_{RST} can be 5 μ s~200 μ s. During this period, no command will be accepted.

Data corruption may happen if there is an on-going internal Erase or Program operation when Reset command is accepted by the device. It is recommended to check the OIP bit in Status Register before issuing the Reset command.

Figure 8 Soft Reset Sequence



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【10.3 读取 JEDEC ID (9Fh)】

该指令兼容 JEDEC SPI 串行存储器标准。

JEDEC ID：Manufacture ID = CDh；Device ID Byte1 = 53h；Device ID Byte2 = 53h。

图9：发送 9Fh，经过 8 个 Dummy Clock 后依次输出 MID、DID Byte1、DID Byte2，均 MSB first。

【10.4 Feature 操作】

10.4.1 Get Feature (0Fh) /Set Feature (1Fh) 用于监视器件状态和修改器件行为。命令带 1-byte feature address，以选择读取或修改的寄存器。OTP、Block Lock 等功能由对应 feature bit 控制。多数 Status 位只读；WEL 由 Write Enable/Disable 命令控制。除特别说明外，Feature 设置会保持到重新上电或再次写入，即使执行 RESET (FFh) 也不改变。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

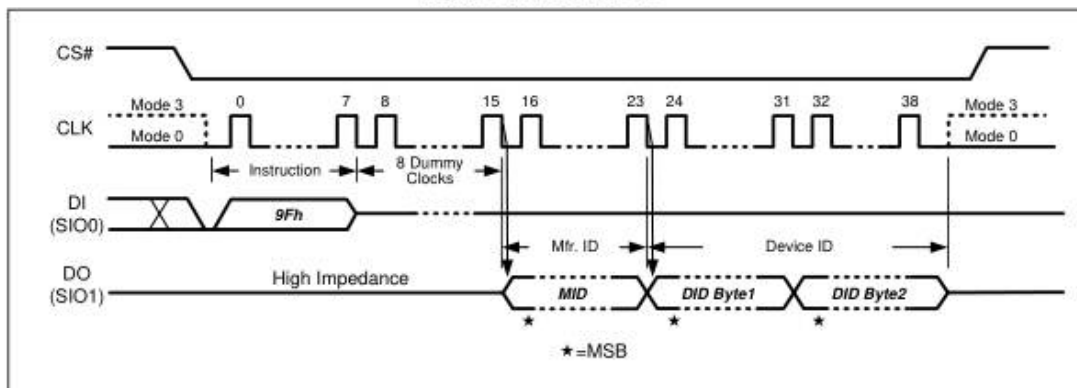
10.3 Read JEDEC ID (9Fh)

The Read JEDEC ID instruction is compatible with the JEDEC standard for SPI compatible serial memories that was adopted in 2003.

Table 16 JEDEC ID

ID		Value
Manufacture ID		CDh
Device ID	Byte 1	53h
	Byte 2	53h

Figure 9 Read JEDEC ID



10.4 Feature Operations

10.4.1 Get Feature (0Fh) and Set Feature (1Fh)

The Get Feature (0Fh) and Set Feature (1Fh) commands are used to monitor the device status and alter the device behavior. These commands use a 1-byte feature address to determine which feature is to be read or modified. Features such as OTP and block locking can be enabled or disabled by setting specific feature bits. The status register is mostly read, except WEL, which is a writable bit with the Write Enable (06h) and Write Disable (04h) command. When a feature is set, it remains active until the device is power cycled or the feature is written to. Unless otherwise specified, once the device is set, it remains set, even if a RESET (FFh) command is issued. Refer to **Status Registers** for detail information.

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【图10 Get Feature / 图11 Set Feature】

Get Feature：0Fh + 1Byte 寄存器地址，随后从 SIO1 输出 1Byte 数据。

Set Feature：1Fh + 1Byte 寄存器地址 + 1Byte 设置数据。

【10.4.2 写使能 WREN（06h）】

WREN 将 WEL 置 1。每次 Page Program、Block Erase 和 OTP 编程/锁定前必须先置 WEL。图12为 06h 写使能时序。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Figure 10 Get Feature

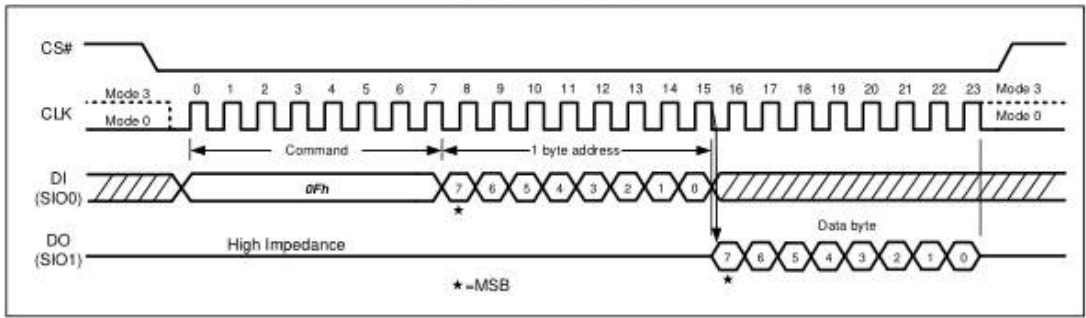
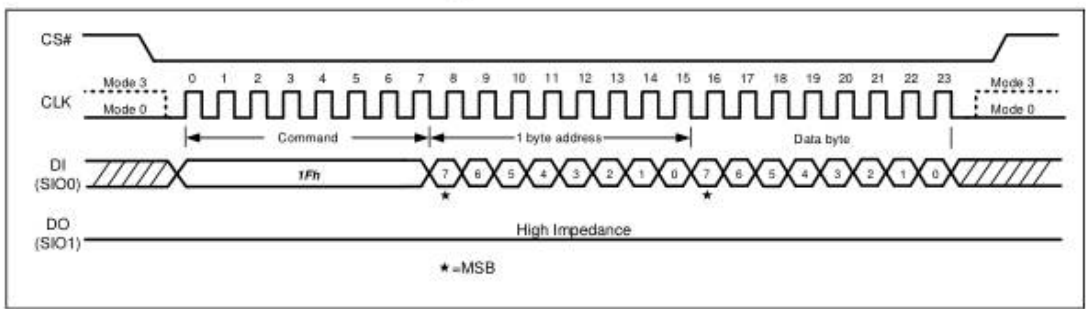


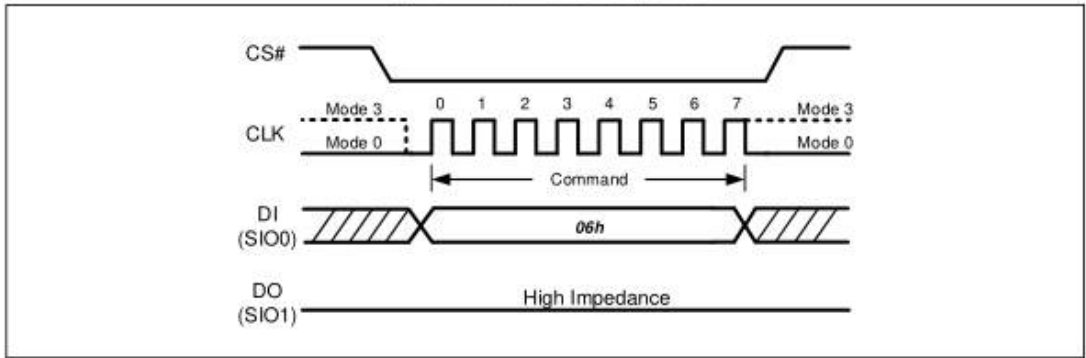
Figure 11 Set Feature



10.4.2 Write Enable (WREN, 06h)

The Write Enable (WREN, 06h) command is for setting Write Enable Latch (WEL) bit. The WEL bit must be set prior to every Page Program, Block Erase and OTP.

Figure 12 Write Enable Sequence



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【10.4.3 写禁止 WRDI (04h)】

WRDI 将 WEL 清 0。上电后以及 Page Program、Block Erase、Reset 等操作完成后，WEL 也会自动清 0。

【10.5 读操作】

支持 Power-on Read：器件上电后自动将 Block0/Page0 从阵列加载到 cache，主机可直接从 cache 读取该页。

【10.5.1 Page Read (13h)】

13h + 24bit 地址将 NAND array 页面传入 cache。CS# 拉高后器件忙 tRD 或 tRD_ECC，可用 Get Feature 轮询 OIP。读取完成后发送 03h/0Bh/3Bh/6Bh 从 cache 输出数据。

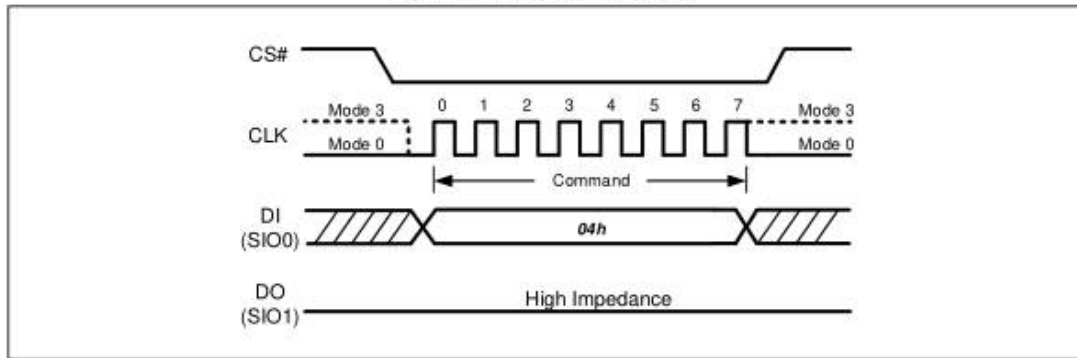
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

10.4.3 Write Disable (WRDI, 04h)

The Write Disable (WRDI, 04h) instruction is to reset Write Enable Latch (WEL) bit. Note that the WEL bit is automatically reset after Power-up and upon completion of the Page Program, Block Erase, and Reset commands.

Figure 13 Write Disable Sequence



10.5 Read Operations

The device supports Power-on Read function, after power up, the device will automatically load the data of the 1st page of 1st block from array to cache. The host micro-controller may directly read the 1st page of 1st block data from the cache buffer.

10.5.1 Page Read (13h)

The page read operation transfers data from array to cache by issuing the Page Read (13h) command followed by the 24-bit address (including the dummy/block/page address).

The device will have a period of time (t_{RD} or t_{RD_ECC}) being busy after the CS# goes high. The Get Feature command may be used to poll the operation status.

After read operation is completed, the Read from Cache (03h or 0Bh), Read from cache (x2) (3Bh), Read from cache (x4) (6Bh) may be issued to fetch the data.

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【图14 Page Read 时序】

13h + 24bit Page Address；等待内部读取完成，可用 0Fh + Status Register 地址获取状态。

【10.5.2 Read From Cache (03h/0Bh)】

在 Page Read 完成后，从 Data Buffer 以指定 Column Address 连续读出一个或多个 Byte。03h 和 0Bh 时序均包含 16bit Column Address 和 8 个 Dummy Clock，然后连续输出 Data Out。

【10.5.3 Read From Cache x2 (3Bh)】

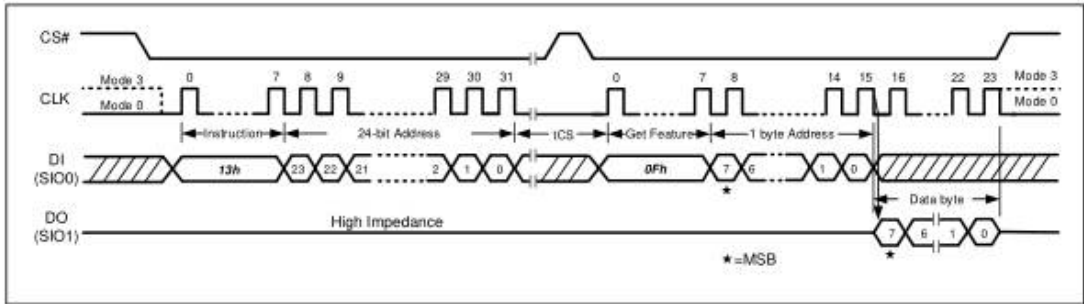
与 03h/0Bh 类似，但数据在 SIO0、SIO1 两条线上并行输出，吞吐率约为标准 SPI 的 2 倍。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Figure 14 Page Read Sequence



10.5.2 Read From Cache (03h or 0Bh)

The Read From Cache command allows one or more data bytes to be sequentially read from the Data Buffer after executing the Read Page command.

Figure 15 Read From Cache (03h) Sequence

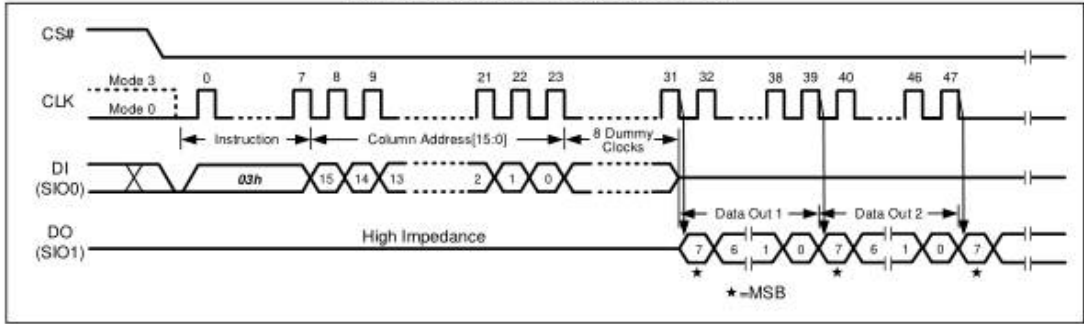
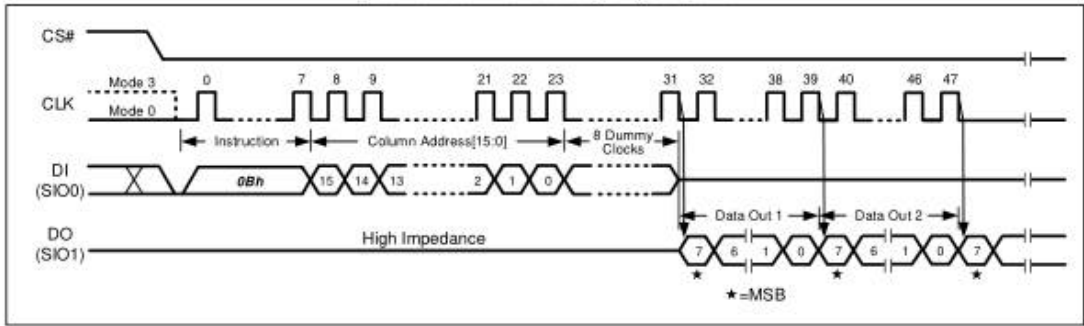


Figure 16 Read From Cache (0Bh) Sequence



10.5.3 Read From Cache x 2 (3Bh)

The Read From Cache x 2 (3Bh) command is similar to the Read From Cache (03h or 0Bh) command

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【10.5.3 x2 读（续）】

3Bh + 16bit Column Address + 8 Dummy Clock, 随后 SIO0/SIO1 以 2bit 并行方式连续输出。

【10.5.4 Read From Cache x4（6Bh）】

与 x2 类似, 但使用 SIO0~SIO3 四条线并行输出, 吞吐率约为标准 SPI 的 4 倍。QE=0 时该命令禁用。

【10.6 编程操作 - 10.6.1 Page Program】

Page Program 可在单页内编程 1 Byte 到整页数据, 典型流程：

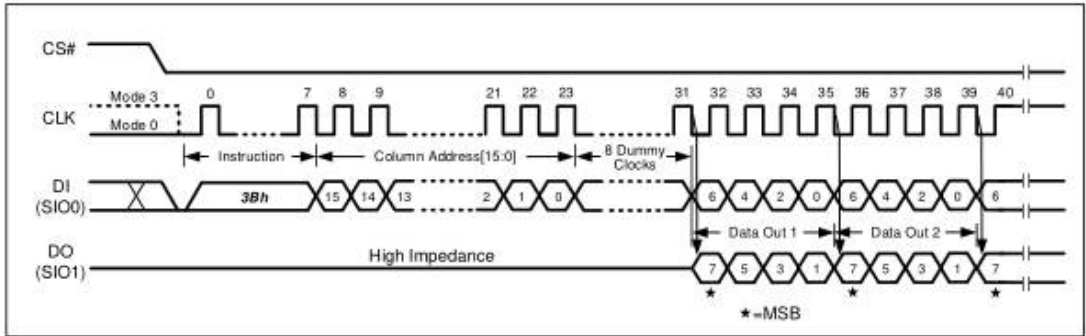
1. Program Data Load（02h）或 x4（32h）；
2. Write Enable（06h）；
3. Program Execute（10h）；
4. Get Feature（0Fh）读取状态。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识, 用于与中文译文交叉核对。

except that data is output on two pins: SIO0 and SIO1. This allows data to be transferred at twice the rate of standard SPI devices.

Figure 17 Read From Cache x2 (3Bh) Sequence

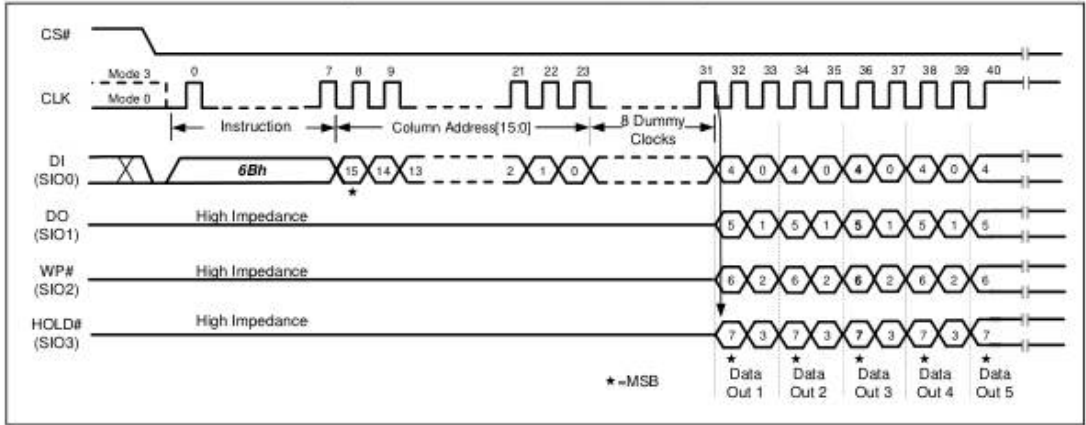


10.5.4 Read From Cache x 4 (6Bh)

The Read From Cache x 4 (6Bh) command is similar to the Read From Cache x 2 (3Bh) command except that data is output on four pins: SIO0, SIO1, SIO2 and SIO3. This allows data to be transferred at four times the rate of standard SPI devices.

When QE bit in the Status Register is set to a 0, this command is disabled.

Figure 18 Read From Cache x4 (6Bh) Sequence



10.6 Program Operations

10.6.1 Page Program

The Page Program operation sequence programs 1 byte to whole page of data within a page. The page

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【10.6.2 Program Data Load (02h) /Random Program Data Load (84h)】
用于把待编程数据装入 Data Buffer。CS# 拉低后发送 02h/84h + 16bit Column Address (仅 CA[12:0] 有效) + 至少 1Byte 数据。
整个数据传输期间 CS# 必须保持低。若输入数据超过 Data Buffer 容量，多余数据被忽略。
随机页编程时必须先执行 Program Data Load，再执行 Random Program Data Load。区别：02h 会将 Data Buffer 中未写入的字节复位为 FFh；84h 只更新命令指定的数据，其余缓存保持原值。
图19给出 02h/84h 时序。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

program sequence is as follows:

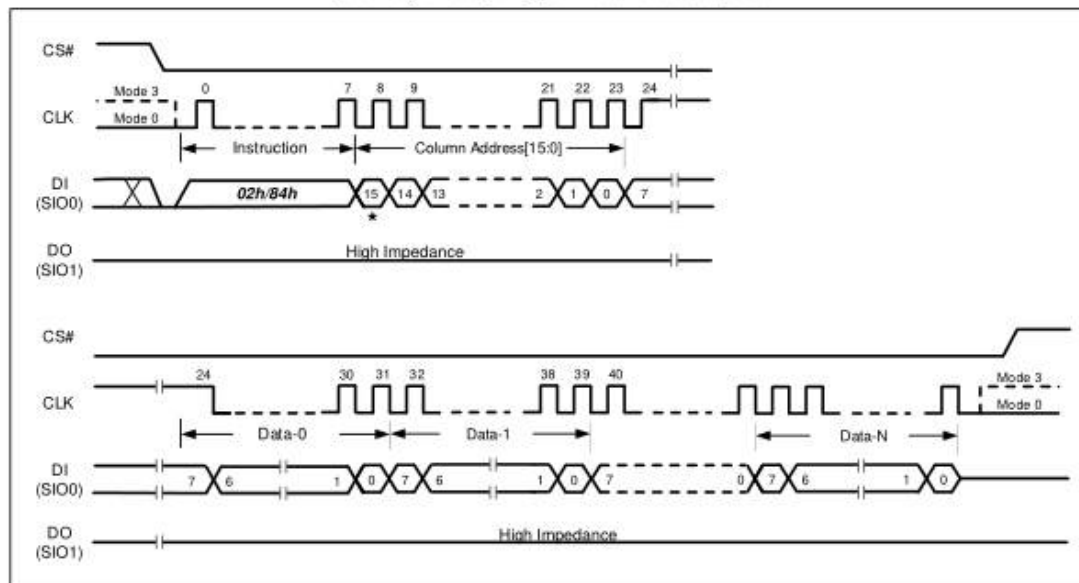
1. Issue Program Data Load (02h) / Program Data Load x4 (32h)
2. Issue Write Enable command (06h)
3. Issue Program Execute command (10h)
4. Issue Get Feature command (0Fh) to read the status

10.6.2 Program Data Load (02h) / Random Program Data Load (84h)

The Program Data Load or Random Program Data Load command is used to load the program data into the data buffer. The command is initiated by driving the CS# pin low then shifting the command code "02h" or "84h" followed by a 16-bit column address (only CA[12:0] is effective) and at least one byte of data into the DI pin. The CS# pin must be held low for the entire length of the instruction while data is being sent to the device. If the number of data bytes sent to the device exceeds the number of data bytes in the Data Buffer, the extra data will be ignored by the device.

The Program Load Data command has to be issued prior to Random Program Load Data command for random page program. The difference is that Program Load Data command will reset the unused data bytes in the Data Buffer to FFh value, while Random Program Load Data command will only update the data bytes that are specified by the command input sequence, the rest of the Data Buffer will remain unchanged.

Figure 19 (Random) Program Data Load Sequence



10.6.3 Program Data Load x 4 (32h) / Random Program Data Load x 4 (34h)

The Program Data Load x 4 and Random Program Data Load x 4 commands are similar to the Program

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【10.6.3 Program Data Load x4 (32h) /Random Program Data Load x4 (34h)】

功能分别对应 02h/84h，但数据由 SIO0~SIO3 四线并行输入，可显著缩短大量数据装载时间。随机页编程时 32h 必须先于 34h；32h 会把未用缓存字节置 FFh，34h 只修改指定字节。QE=0 时所有 Quad SPI 指令禁用。

【10.6.4 Program Execute (10h)】

将 Data Buffer 内容写入命令指定的物理 NAND 页。执行前必须先 WREN (06h) 使 WEL=1。发送 10h + 7 Dummy Clock + 17bit Page Address。CS# 拉高后启动内部自定时编程，持续 tPROG 或 tPROG_ECC。编程期间可用 0Fh 轮询 OIP；完成后 OIP=0 且 WEL 自动清 0。若目标页受 TB/BP3-0 保护，Program Execute 不执行。每个页只允许 1 次 partial page program。

原页图示参考

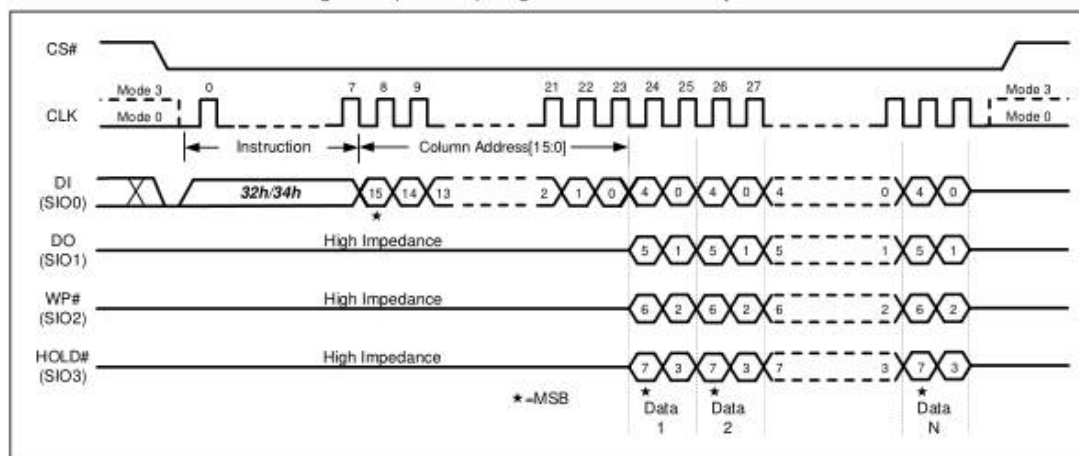
以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Load Data and Random Program Load Data, the only difference is that "x4" commands will input the data bytes from all four IO pins instead of the single DI pin. This method will significantly shorten the data input time when a large amount of data needs to be loaded into the Data Buffer.

The Program Data Load x 4 command has to be issued prior to Random Program Data Load x 4 command for random page program. The difference is that Program Data Load x 4 command will reset the unused data bytes in the Data Buffer to FFh value, while Random Program Data Load x 4 instruction will only update the data bytes that are specified by the command input sequence, the rest of the Data Buffer will remain unchanged.

When QE bit in the Status Register is set to 0, all Quad SPI instructions are disabled.

Figure 20 (Random) Program Data Load x4 Sequence



10.6.4 Program Execute (10h)

The Program Execute command will program the Data Buffer content into the physical memory page that is specified in the command. Prior to performing the Program Execute operation, a Write Enable (06h) command must be issued to set the WEL bit.

The Program Execute command is initiated by driving the CS# pin low then shifting the instruction code "10h" followed by 7-bit dummy clocks and the 17-bit Page Address into the DI pin.

After CS# is driven high to complete the instruction cycle, the self-timed Program Execute instruction will commence for time duration of **tPROG** or **tPROG_{ECC}**. While the Program Execute cycle is in progress, the Get Feature command (0Fh) may be used for checking the status of the OIP bit. The OIP bit is a 1 during the Program Execute cycle and becomes a 0 when the cycle is finished and the device is ready to accept other instructions again. After the Program Execute cycle has finished, the Write Enable Latch (WEL) bit in the Status Register is cleared to 0. The Program Execute command will not be executed if the addressed page is protected by the Block Protect (TB, BP3, BP2, BP1, and BP0) bits. Only 1 partial

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【10.6.4 Program Execute（续）】

同一块内必须按页地址从低到高顺序编程，不允许乱序页地址编程。图21给出 10h + Page Address 时序。

【10.6.5 内部数据搬移】

用于以现有页面数据为基础，在器件内部编程/替换目标页：

1. Page Read（13h）把源页读入 cache；
2. 可选：Program Load Random Data（84h/34h）修改部分字节；
3. WREN（06h）；
4. Program Execute（10h）写入目标页；
5. Get Feature（0Fh）检查状态。

【10.7 Block Erase】

Block Erase 将指定块全部置为擦除态 1（FFh）。执行前必须 WEL=1。命令为 D8h + 7 Dummy Clock + 17bit Page Address。

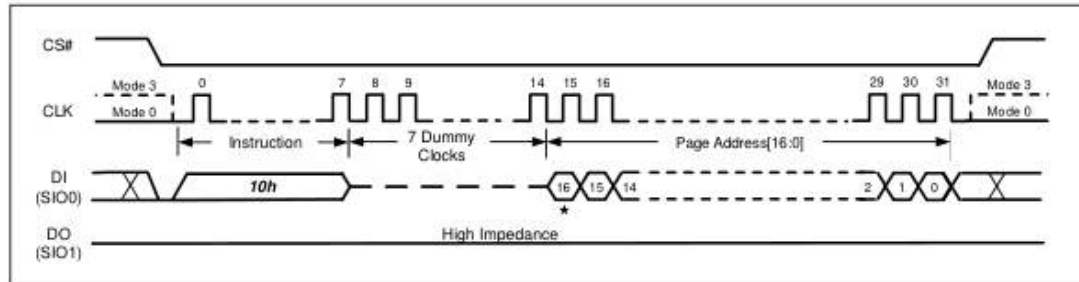
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

page program time is allowed on every single page.

The pages within the block have to be programmed sequentially from the lower order page address to the higher order page address within the block. Programming pages out of sequence is prohibited.

Figure 21 Program Execute Sequence



10.6.5 Internal Data Move

The Internal Data Move command sequence programs or replaces data in a page with existing data. The sequence is as follows:

1. Issue Page Read command (13h)
2. Program Load Random Data (Optional)
3. Issue Write Enable command (06h)
4. Issue Program Execute command (10h)
5. Issue Get Feature command (0Fh) to read the status

Prior to performing an internal data move operation, the target page content must be read out into the cache register by issuing a Page Read (13h) command. One or more Program Load Random Data (84h/34h) command can be issued, if user wants to update bytes of data in the page. After the data is loaded, the Write Enable command (06h) and the Program Execute (10h) command can be issued to start the program operation.

10.7 Block Erase Operations

The Block Erase instruction sets all memory within a specified block to the erased state of all 1s (FFh). A Write Enable command must be executed before the device will accept the Block Erase Instruction (Status Register bit WEL must equal 1). The command is initiated by driving the CS# pin low and shifting the command code "D8h" followed by 7-bit dummy clocks and the 17-bit page address.

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【图22 Block Erase 时序】

D8h + 7 Dummy Clock + Page Address[16:0], CS# 拉高后开始块擦除。

【10.8 UID / Parameter / OTP 页】

主阵列以外还包括：

PA 0_00_00h：Unique ID Page，工厂编程，只读，32Byte x16。

PA 0_00_01h：Parameter Page，工厂编程，只读，256Byte x3。

PA 0_00_02h：OTP Page0，仅可编程，可 OTP Lock，完整页。

中间地址：OTP Page1~60。

PA 0_00_3Fh：OTP Page61。

Unique ID 页保存 16 份 UID + 按位反码，每份 32Byte，前 16Byte 与后 16Byte 互补；若 UID 与其反码 XOR 全为 1，则该 UID 有效。

Parameter Page 至少保存 3 份完全相同的 256Byte 参数数据。

访问这些附加页前必须把 OTP-E=1；UID/Parameter 可读，未锁定 OTP 可读写。返回主阵列操作时将 OTP-E=0。

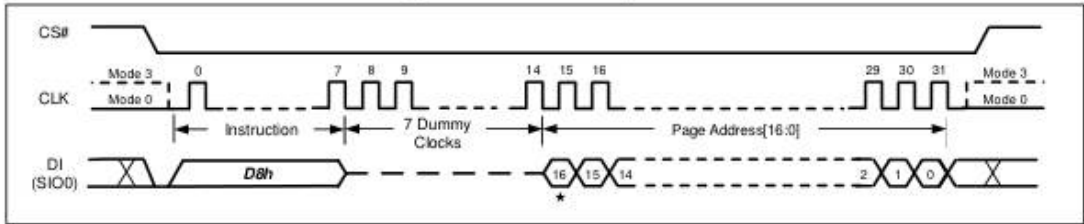
10.8.1 读取流程：Set Feature 令 OTP-E=1 -> Page Read (13h) 指定上述地址 -> Get Feature 查看状态 -> Read From Cache (03h/0Bh/3Bh/6Bh) 取数。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Figure 22 Block Erase Sequence



10.8 UID / Parameter / OTP Pages

In addition to the main memory array, the device has one Unique ID Page, one Parameter Page, and sixty-two OTP Pages.

Table 17 UID / Parameter / OTP Pages

Page Address PA[16:0]	Page Name	Descriptions	Data Length
0_00_00h	Unique ID Page	Factory programmed, Read Only	32-Byte x 16
0_00_01h	Parameter Page	Factory programmed, Read Only	256-Byte x 3
0_00_02h	OTP Page [0]	Program Only, OTP lockable	Full Page
...	OTP Pages [1:60]	Program Only, OTP lockable	Full Page
0_00_3Fh	OTP Page [61]	Program Only, OTP lockable	Full Page

Unique ID Page: To accommodate robust retrieval of the UID in the case of bit errors, sixteen copies (each copy has 32 bytes) of the UID and the corresponding complement are stored. On each 32-byte, the first 16-byte and following 16-byte are complementary. If the XOR of the UID and its bit-wise complement is all ones, then the UID is valid.

Parameter Page: Contains at least three identical copies of the 256-Byte Parameter Data.

To access these additional data pages, the OTP-E bit in Configuration Register must be set to "1" first. Then, Read operations can be performed on Unique ID and Parameter Pages, Read and Program operations can be performed on the OTP pages if it's not already locked. To return to the main memory array operation, OTP-E bit needs to be set to 0.

10.8.1 Read UID / Parameter / OTP Pages

The Read UID / Parameter / OTP pages sequence is as follows:

- 1. Issue Set Feature command (1Fh) to set OTP-E=1.
- 2. Issue Page Read command (13h) with address shown in the table above.
- 3. Issue Get Feature command (0Fh) to read the status.
- 4. Issue Read from cache command (03h/0Bh/3Bh/6Bh) to read data.

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【10.8.1 注意】

读取 OTP 页时可启用内部 ECC 以提高数据完整性；读取 UID/Parameter Page 时，芯片自动禁用内部 ECC。

【10.8.2 OTP 页编程与锁定】

62 个 OTP 页共提供 4KB x62 的一次性安全存储空间，出厂为擦除态，只能从 1 编程为 0；OTP-L 锁定后永久禁止修改。
OTP 编程：1) OTP-E=1；2) WREN；3) Program Data Load + Program Execute；4) Get Feature 检查状态。ECC 开启时 Program Execute 会计算 ECC。

OTP 永久锁定：1) Set Feature 同时 OTP-E=1、OTP-L=1；2) WREN；3) Program Execute (10h, Page Address don't care)；4) Get Feature 检查；5) Set Feature 令 OTP-E=0 返回主阵列。

【10.8.3 参数页数据定义（前半）】

Byte0~3：参数页签名 ASCII "ONFI" = 4F 4E 46 49h。

4~5：Revision Number = 0000h。

6~31：保留，全 00h。

32~43：厂商 12 ASCII 字符，内容为 "FORESEE" 后补空格。

44~63：器件型号 20 ASCII 字符，内容为 "F35SQB004G" 后补空格。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Note:

- (1) For OTP pages, Internal ECC can be enabled for the OTP page read operations to ensure the data integrity.
- (2) When reading UID / Parameter page, Internal ECC is disabled by the chip.

10.8.2 Program OTP Pages and OTP Lock Operation

OTP pages provide the additional space (4K-Byte x 62) to store important data or security information that can be locked to prevent further modification in the field. These OTP pages are in an erased state set in the factory, and can only be programmed (change data from "1" to "0") until being locked by OTP-L bit in the Configuration Register.

The Program OTP Pages sequence is as follows:

1. Issue Set Feature command (1Fh) to set OTP-E=1
2. Issue WREN command (06h) to set WEL bit
3. Issue Program Data Load and Program Execute command
4. Issue Get Feature command (0Fh) to read the status.

When ECC is enabled, ECC calculation will be performed during Program Execute.

Once the OTP pages are correctly programmed, OTP-L bit can be used to permanently lock these pages so that no further modification is possible.

The OTP Lock sequence is as follows:

1. Issue Set Feature command (1Fh) to set OTP-E=1 and OTP-L=1
2. Issue WREN command (06h) to set WEL bit
3. Issue Program Execute command (10h), page address is "don't care"
4. Issue Get Feature command (0Fh) to read the status.
5. Issue Set Feature command (1Fh) to set OTP-E=0, return to the main memory array operation.

10.8.3 Parameter Page Data Definition

Table 18 Parameter Definition

Byte Number	Descriptions	Values
0~3	Parameter Page Signature, "ONFI" ASCII characters	4Fh 4Eh 46h 49h
4~5	Revision Number	00h 00h
6~31	Reserved (0)	all 00h
32~43	Device manufacturer , 12 ASCII characters	46h 4Fh 52h 45h 53h 45h 45h 20h 20h 20h 20h 20h
44~63	Device Model, 20 ASCII characters	46h 33h 35h 53h 51h 42h 30h 30h 34h 47h 20h 20h 20h 20h 20h 20h 20h 20h 20h 20h

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【参数页数据定义（续）】

64 JEDEC MID=CDh；65~66 Date Code=0000h；67~79 保留 0。
80~83 每页 Data Bytes=00 10 00 00h（4096）。
84~85 每页 Spare Bytes=00 01h（256）。
86~89 每 Partial Page Data Bytes=00 02 00 00h（512）。
90~91 每 Partial Page Spare Bytes=20 00h（32）。
92~95 每块页数=40 00 00 00h（64）。
96~99 每 Logic Unit 块数=00 08 00 00h（2048）。
100 Logic Unit 数=01h；102 每 Cell bit 数=01h。
103~104 每 Logic Unit 最大坏块数=28 00h；105~106 Block Endurance=06 04h；107 Target 开始处保证有效块数=01h；
108~109 保证有效块耐久值=01 03h；110 每页允许 Program 次数=01h。
111 Partial Programming Attributes=00h；112 ECC 可纠正 bit 数字段=00h；113~127 保留 0。
128 最大 I/O Pin Capacitance=08h；129~132 保留。
133~134 tPROG 最大页编程时间(us)=BC 02h；135~136 tERS 最大块擦除时间(us)=10 27h；137~138 tRD 最大页读时间(us)=73 00h。
139~163 保留；164~165 Vendor Specific Revision=0000h；166~253 Vendor Specific 全 00h；254~255 Integrity CRC=07 CDh。
256~511、512~767：Byte0~255 的冗余副本；768+：其他冗余参数页。
Integrity CRC 用于确认参数页正确传输到主机，计算规则参照 ONFI 1.0，16bit 生成多项式 $G(X)=X^{16}+X^{15}+X^2+1$ 。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Byte Number	Descriptions	Values
64	JEDEC MID	CDh
65-66	Date Code	00h 00h
67-79	Reserved (0)	all 00h
80-83	Number of Data Bytes per Page	00h 10h 00h 00h
84-85	Number of Spare Bytes per Page	00h 01h
86-89	Number of Data Bytes per Partial Page	00h 02h 00h 00h
90-91	Number of Spare Bytes per Partial Page	20h 00h
92-95	Number of Pages per Block	40h 00h 00h 00h
96-99	Number of Block per Logic Unit	00h 08h 00h 00h
100	Number of Logic Units	01h
101	Reserved (0)	00h
102	Number of Bits per Cell	01h
103-104	Bad Blocks Maximum per Logic Unit	28h 00h
105-106	Block Endurance	06h 04h
107	Guaranteed Valid Blocks at Beginning of Target	01h
108-109	Block Endurance for Guaranteed Valid Blocks	01h 03h
110	Number of Programs per Page	01h
111	Partial Programming Attributes b5-b7 reserved (0) b4 1 = partial page layout is partial page data followed by partial page spare b1-b3 reserved (0) b0 1 = partial page programming has constraints	00h
112	Number of ECC Bits Correctability	00h
113-127	Reserved (0)	all 00h
128	I/O Pin Capacitance, Maximum	08h
129-132	Reserved (0)	all 00h
133-134	t _{PROG} Maximum Page Program Time (us)	BCh 02h
135-136	t _{ERS} Maximum Block Erase Time (us)	10h 27h
137-138	t _{RD} Maximum Page read Time (us)	73h 00h
139-163	Reserved (0)	all 00h
164-165	Vendor Specific Revision Number	00h 00h
166-253	Vendor Specific	all 00h
254-255	Integrity CRC	07h CDh
256-511	Value of Bytes 0-255	
512-767	Value of Bytes 0-255	
768+	Additional Redundant Parameter Pages	

Note:

- (1) The Integrity CRC (Cycling Redundancy Check) field is used to verify that the contents of the parameters page were transferred correctly to the host. Please refer to ONFI 1.0 specifications for details. The CRC shall be calculated using the following 16-bit generator polynomial: $G(X) = X^{16} + X^{15} + X^2 + 1$

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【11 软件算法】

11.1 初始无效块：初始无效块包含一个或多个出厂即不保证可靠性的无效 bit。带初始无效块的器件与全良块器件具有相同质量等级和电气特性，坏块不影响其他良块性能。系统必须通过地址映射屏蔽初始无效块。Block0（地址 00h）出货时保证为良块。

有效块数 NVB：Min 2008，Max 2048 Blocks。

11.2 识别初始无效块：除写入坏块标记的位置外，器件出货时均为 FFh。每个初始坏块在第 1 页或第 2 页 spare 区的第 1 Byte 标记为非 FFh。由于该标记多数情况下可被擦除，一旦擦除就无法恢复，因此系统必须在初始阶段读取原始坏块标记并建立 Initial Invalid Block Table，禁止有意擦除原始坏块信息。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



11 Software Algorithm

11.1 Initial Invalid Block(s)

Initial invalid blocks are defined as blocks that contain one or more initial invalid bits whose reliability is not guaranteed. Devices with initial invalid block(s) have the same quality level as devices with all valid blocks and have the same electrical characteristics. An initial invalid block(s) does not affect the performance of valid block(s). The system design must be able to mask out the initial invalid block(s) via address mapping. The 1st block, which is placed on 00h block address, is guaranteed to be a valid block at the time of shipment.

Table 19 Valid Block Number

Parameter	Symbol	Min	Max	Unit
Valid block number	N _{VB}	2008	2048	Blocks

11.2 Identifying Initial Invalid Block(s)

All device locations are erased (FFh) except locations where the initial invalid block(s) information is written prior to shipping. All initial invalid blocks are marked with non-FFh at the first byte of spare area on the 1st or 2nd page. Since the initial invalid block information is also erasable in most cases, it is impossible to recover the information once it has been erased. Therefore, the system must be able to recognize the initial invalid block(s) based on the original initial invalid block information and create the initial invalid block table via the suggested flow (Figure 23). Any intentional erasure of the original initial invalid block information is prohibited.

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【图23 建立初始无效块表流程】

从 Block Address=0 开始；检查每个块第 1 页和第 2 页 Column Address 4096 处是否为 FFh。若不是 FFh，则创建/更新初始无效块表；然后地址递增，直到最后一个 Block。

【11.3 操作错误】

NAND Flash 使用寿命期间可能产生新的坏块。系统应针对以下失败模式实施可靠性处理：

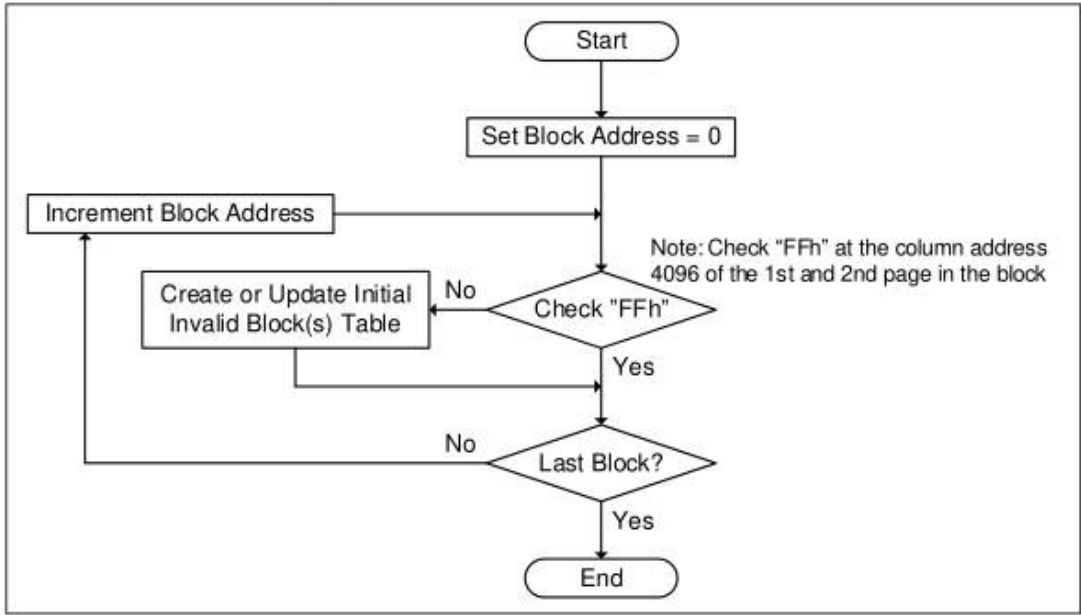
- Erase：擦除后读状态；失败则进行 Block Replacement。
- Program：编程后读状态；失败则进行 Block Replacement。
- Read：使用 ECC 校验并纠错。

Program Status Fail 通常不影响同一块其他页，可寻找空闲已擦除块，用 page-sized buffer 重写当前目标数据并复制原块其余页。对于可由 ECC 纠正的读错误，建议暂不更换块，以提高空间利用率；此类可恢复块不计入新增坏块率。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Figure 23 Flow to Create Initial Invalid Block Table



11.3 Error in Operation

Within its life time, additional invalid blocks may develop with NAND Flash memory. The following possible failure modes should be considered to implement a highly reliable system. In the case of status read failure after erase or program, block replacement should be done. Because program status fail during a page program does not affect the data of the other pages in the same block, block replacement can be executed with a page-sized buffer by finding an erased empty block and reprogramming the current target data and copying the rest of the replaced block. In case of Read, ECC must be employed. To improve the efficiency of memory space, it is recommended that the blocks with correctable bits error reclaimed by ECC don't need to be replaced. The said additional block failure rate does not include those reclaimed blocks.

Table 20 Failure Modes

Operation	Detection and recommended procedure
Erase	Status read after erase → Block Replacement
Program	Status read after program → Block Replacement
Read	Verify ECC → ECC correction

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【图24 坏块替换】

示意原坏块中的有效页数据与当前待写数据迁移到新的擦除块，坏块被加入坏块管理表，不再使用。

【11.4 内部 ECC】

每个 ECC sector 最多可检测/纠正 8bit 错误。1 sector = 512Byte main + 32Byte spare。内部 ECC 默认开启，可通过 Set Feature 改变 ECC-E，并用 Get Feature 确认状态。

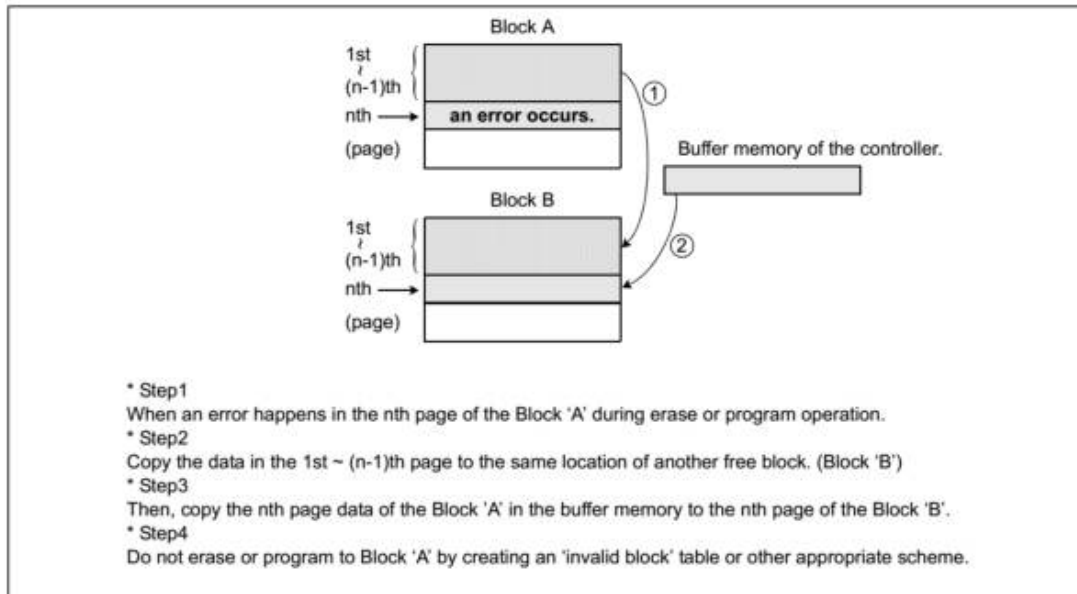
ECC 开启时：

- Main 与 Spare 数据均受保护，具体范围由 ECC-M 决定。
- Program 时计算 ECC code 并存入额外 spare 区。
- Read 时等待 tRD_ECC 完成后，通过 Get Feature 读取 ECCS 判断纠错是否成功；还可读取 Sector ECC Status 获得每个 sector 更详细的错误数量。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Figure 24 Bad Block Replacement



11.4 Internal ECC

The internal ECC logic may detect and correct no more than 8-bit error in each ECC sector. An ECC sector is composed by a main area (512 Byte) and a spare area (32 Byte). The default state of the internal ECC is enabled. It is operated by the Set Feature operation to enable or disable internal ECC, and then check the internal ECC state by Get Feature operation.

The internal ECC is enabled by using Set Feature command (1Fh) to set ECC-E. To disable the internal ECC can be done by using the Set Feature command (1Fh) to clear ECC-E.

When the internal ECC is enabled:

- Data in main area and spare area are protected. ECC protection range is depended on ECC-M bit config, refer to **ECC Mode Bit (ECC-M)** for detailed information.
- During a Program operation, an ECC code is calculated and stored in the additional spare area.
- During a Read operation, after the data transfer time (t_{RD_ECC}) is completed, a Get Feature command should be issued to check the ECC status bits which indicate whether or not the error correction was successful. Please refer to **Status Register**. Furthermore, user could check the detailed bits error information for each ECC sector by Get Feature operation. See **Sector ECC Status Register** for details.

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【11.5 编程操作寻址】

一个块内的页必须从较低页地址向较高页地址连续编程。这里的 LSB page 指本次需要编程的一组页中的起始页，不要求一定是该块 Page0。禁止随机页地址编程。图25给出允许的页地址递增方式。

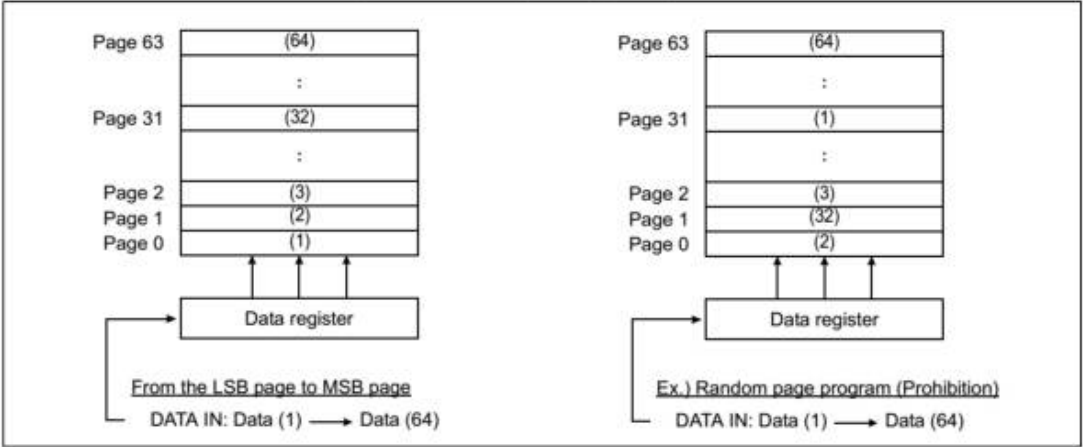
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

11.5 Addressing for Program Operation

Within a block, the pages must be programmed consecutively from the LSB (least significant bit) page of the block to the MSB (most significant bit) pages of the block. The LSB page is defined as the start page among the pages to be programmed, does not need to be page 0 in the block. Random page address programming is prohibited.

Figure 25 Addressing for Program Operation



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【12 电气特性】

12.1 绝对最大额定值：

VCC 电源电压 -0.6~+4.6V；任意引脚电压 VIO -0.6~VCC+0.4V；偏置温度 TBIAS -40~+125 °C；存储温度 TSTG -65~+150 °C；I/O 短路输出电流 IOS=5mA。

瞬态期间 I/O 可短时下冲至 -2.0V (<30ns)，或上冲至 VCC+2.0V (<20ns)。超过绝对最大额定值可能造成永久损坏；长期暴露在极限条件会影响可靠性，正常工作必须遵守推荐工作条件。

12.2 工作范围：VCC 2.7~3.6V；Industrial TA=-40~+85 °C；Industrial Plus TA=-40~+105 °C。

12.3 上电/掉电时序：VCC(min) 到允许读 Status Register 的 tVSL 最小 200us；器件完全可访问前延时 tPUW 最小 1ms。参数为特性化值。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

12 Electrical Characteristics

12.1 Absolute Maximum Ratings

Table 21 Absolute Maximum Rating

Parameters	Symbol	Range	Unit
Supply Voltage	V _{CC}	-0.6 to +4.6	V
Voltage Applied to Any Pin	V _{IO}	-0.6 to V _{CC} +0.4	V
Temperature under Bias	T _{BIAS}	-40 to +125	°C
Storage Temperature	T _{STG}	-65 to +150	°C
Short circuit output current, I/Os	I _{OS}	5	mA

Note:

- (1) Minimum DC voltage is -0.6V on input/output pins. During transitions, this level may undershoot to -2.0V for periods <30ns. Maximum DC voltage on input/output pins is V_{CC}+0.3V which, during transitions, may overshoot to V_{CC}+2.0V for periods <20ns.
- (2) Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

12.2 Operating Ranges

Table 22 Operating Ranges

Parameters	Symbol	Conditions	Min	Max	Unit
Supply Voltage	V _{CC}		2.7	3.6	V
Ambient Temperature	T _A	Industrial	-40	+85	°C
		Industrial plus	-40	+105	°C

12.3 Power-up and Power-down Timing Requirements

Table 23 Power-up Timing

Parameters	Symbol	Min	Max	Unit
V _{CC} (min) to read Status Register is allowed	t _{VSL}	200		μs
Time delay before device fully accessible	t _{PUW}	1		ms

Note:

- (1) These parameters are characterized only.

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【图26 上电时序】

VCC 上升越过 VCC(min) 后等待 tVSL，才允许 Read Status Register；再满足 tPUW 后器件完全可访问。

图27：VCC 上升/下降期间，CS# 必须跟随 VCC，避免错误命令。

【12.4 引脚电容】

I/O 电容 CIO Max 8pF；输入电容 CIN Max 8pF。

测试条件：TA=25 ° C、F=1MHz、VIN=0V、VCC=3V；这些参数仅为特性化值。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Figure 26 Power-up Timing

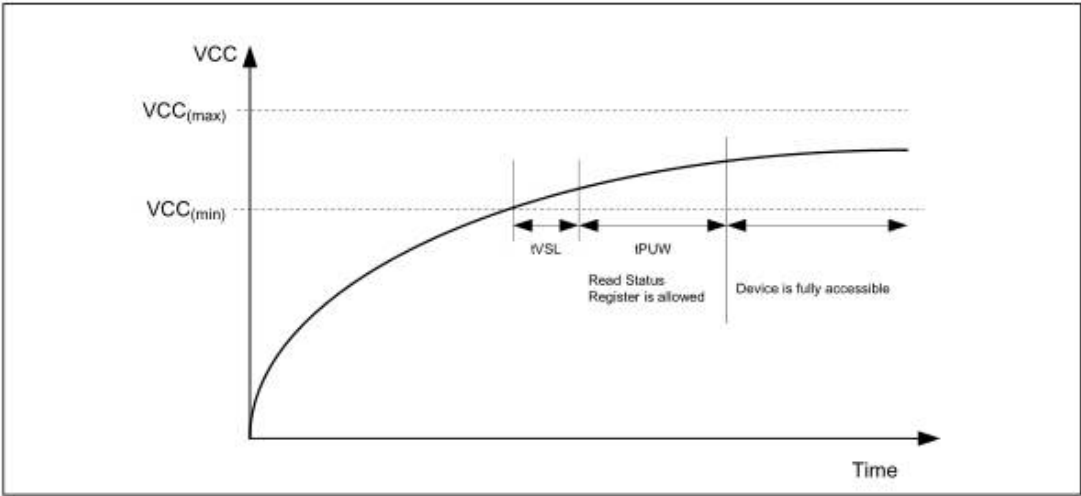
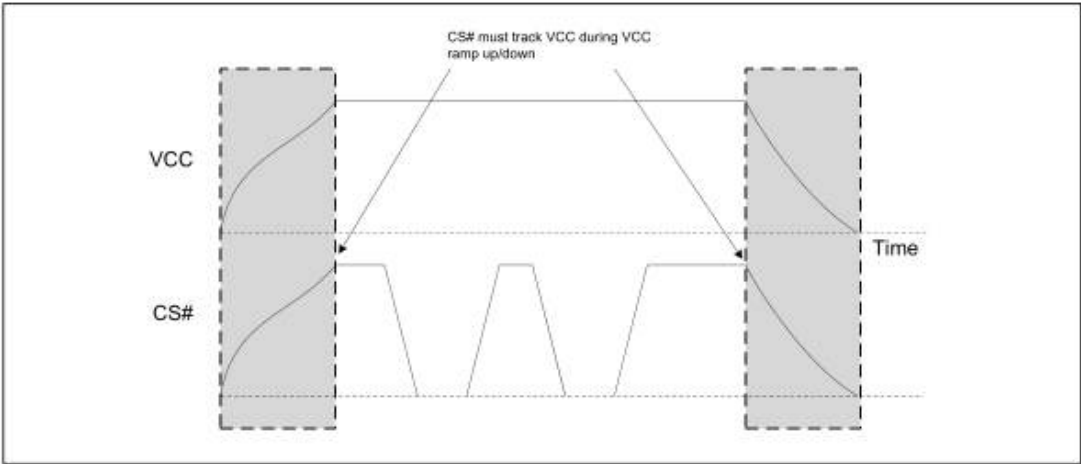


Figure 27 Power-up and Power-down Requirements



12.4 Pin Capacitance

Table 24 Pin Capacitance

Parameters	Symbol	Min	Max	Unit
Input / Output Capacitance	C _{IO}		8	pF
Input Capacitance	C _{IN}		8	pF

Note:

- (1) Test conditions: TA=25°C, F=1MHz, V_{IN}=0V, V_{CC}=3V
- (2) These parameters are characterized only.

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【12.5 直流电气特性】

适用条件（除非另有说明）：TA=-40~+85 °C，VCC=2.7~3.6V。

Standby Current ICC1：CS#=VCC、VIN=GND/VCC，Typ 10uA，Max 50uA。

Page Read Current ICC2：Typ 10mA，Max 25mA。

Program Current ICC3：Typ 15mA，Max 25mA。

Erase Current ICC4：Typ 15mA，Max 25mA。

Input Leakage ILI：Max +/-2uA；Output Leakage ILO：Max +/-2uA。

VIL：Min -0.3V，Max 0.2VCC；VIH：Min 0.8VCC，Max VCC+0.3V。

VOL：IOL=2.1mA，Max 0.4V；VOH：IOH=-400uA，Min 2.4V。

【12.6 AC 测量条件】

CL Max 30pF；输入上升/下降时间 tR/tF Max 2.5ns；输入脉冲 0~VCC；输入/输出时序参考电平 0.5VCC。

【12.7 AC 特性（前半）】

Clock frequency FR Max 166MHz；tCLH/tCLL Min 2.7ns；Clock Rise/Fall Slew tCLCH/tCHCL Min 0.1V/ns；Data In Setup tDVCH Min 2ns。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

12.5 DC Electrical Characteristics

Table 25 DC Electrical Characteristics

Parameters	Symbol	Conditions	SPEC ⁽¹⁾			Unit
			Min	Typ	Max	
Standby Current	I _{CC1}	CS# = V _{CC} , V _{IN} = GND or V _{CC}		10	50	μA
Page Read Current	I _{CC2}			10	25	mA
Program Current	I _{CC3}			15	25	mA
Erase Current	I _{CC4}			15	25	mA
Input Leakage Current	I _{LI}				±2	μA
Output Leakage Current	I _{LO}				±2	μA
Input Low Voltage	V _{IL}		-0.3		0.2V _{CC}	V
Input High Voltage	V _{IH}		0.8V _{CC}		V _{CC} +0.3	V
Output Low Voltage	V _{OL}	I _{OL} = 2.1mA			0.4	V
Output High Voltage	V _{OH}	I _{OH} = -400μA	2.4			V

Note:

(1) Applicable over recommended operating range from: T_A = -40°C to +85°C, V_{CC} = 2.7V to 3.6V, unless otherwise noted.

12.6 AC Measurement Conditions

Table 26 AC Measurement Conditions

Parameters	Symbol	Min	Max	Unit
Load Capacitance	C _L		30	pF
Input Rise Time	t _{RI}		2.5	ns
Input Fall Time	t _{FI}		2.5	ns
Input Pulse Voltages	V _{IN}	0 to V _{CC}		V
Input Timing Reference Voltages	I _N	0.5V _{CC}		V
Output Timing Reference Voltages	O _{UT}	0.5V _{CC}		V

12.7 AC Electrical Characteristics

Table 27 AC Electrical Characteristics

Parameters	Symbol	SPEC ⁽¹⁾			Unit
		Min	Typ	Max	
Clock frequency	F _R			166	MHz
Clock High, Low Time for all commands	t _{CUH} , t _{CLL} ⁽²⁾	2.7			ns
Clock Rise Time	t _{CLCH} ⁽³⁾	0.1			V/ns
Clock Fall Time	t _{CHCL} ⁽³⁾	0.1			V/ns
Data In Setup Time	t _{DVCH}	2			ns

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【12.7 AC 特性（续）】

Data In Hold tCHDX Min 3ns ; Clock Low to Output Valid tCLQV Max 5ns ; Output Hold tCLQX Min 1ns ; Output Disable tSHQZ Max 20ns。

CS# active Setup/Hold tSLCH/tCHSH Min 5ns ; CS# non-active Setup/Hold tSHCH/tCHSL Min 5ns ; CS# deselect tSHSL Min 20ns。

HOLD# active Setup/Hold tHLCH/tCHHH Min 5ns ; non-active Setup/Hold tHHCH/tCHHL Min 5ns ; HOLD# to Output Low-Z tHHQX Max 15ns ; High-Z tHLQZ Max 15ns。

WP# Setup before CS# Low tWHSL Min 20ns ; WP# Hold after CS# Low tSHWL Min 100ns。

Device Reset Time tRST (Read/Program/Erase) Max 5/20/200us。

条件：TA=-40~+85 ° C、VCC=2.7~3.6V；tCLH+tCLL ≤ 1/FR；标注项为特性化参数。

【12.8 读/编程/擦除特性】

tRD Data Transfer Cell->Data Register Max 25us。

tRD_ECC (ECC 开启) Typ 95us, Max 105us。

tPROG Typ 350us, Max 770us。

tPROG_ECC Typ 450us, Max 830us。

tERS Block Erase Typ 2ms, Max 10ms。

NOP 每页 Partial Program 次数 Max 1 cycle。

典型编程时间定义为 3.3V、25 ° C 下，超过 50% 全部页面可在该时间内完成编程。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Parameters	Symbol	SPEC ⁽¹⁾			Unit
		Min	Typ	Max	
Data In Hold Time	t _{CHDX}	3			ns
Clock Low to Output Valid	t _{CLOV}			5	ns
Output Hold Time	t _{CLOX}	1			ns
Output Disable Time	t _{SHOZ} ⁽³⁾			20	ns
CS# active Setup Time	t _{SLCH}	5			ns
CS# active Hold Time	t _{CHSH}	5			ns
CS# non-active Setup Time	t _{SHCH}	5			ns
CS# non-active Hold Time	t _{CHSL}	5			ns
CS# deselect Time	t _{SHSL}	20			ns
HOLD# active Setup Time	t _{HLCH}	5			ns
HOLD# active Hold Time	t _{CHHH}	5			ns
HOLD# non-active Setup Time	t _{HHCH}	5			ns
HOLD# non-active Hold Time	t _{CHHL}	5			ns
HOLD# to Output Low-Z	t _{HHQX} ⁽³⁾			15	ns
HOLD# to Output High-Z	t _{HLQZ} ⁽³⁾			15	ns
WP# Setup Time before CS# Low	t _{WHSL}	20			ns
WP# Hold Time after CS# Low	t _{SHWL}	100			ns
Device reset time (Read/Program/Erase)	t _{RST} ⁽³⁾			5/20/200	μs

Note:

(1) Applicable over recommended operating range from: T_A = -40°C to +85°C, VCC= 2.7V to 3.6V, unless otherwise noted.(2) t_{CLH} + t_{CLL} ≤ 1/F_R.

(3) These parameters are characterized only.

12.8 Read / Program / Erase Characteristics

Table 28 Read / Program / Erase Characteristics

Parameters	Symbol	SPEC ⁽¹⁾			Unit
		Min	Typ	Max	
Data Transfer from Cell to Data Register	t _{RD}			25	μs
Data Transfer from Cell to Data Register with internal ECC enabled	t _{RD_ECC}		95	105	μs
Program Time	t _{PROG} ⁽²⁾		350	770	μs
Program Time with internal ECC enabled	t _{PROG_ECC} ⁽²⁾		450	830	μs
Block Erase Time	t _{ERS}		2	10	ms
Number of Partial Program Cycles	NOP			1	cycles

Note:

(1) Applicable over recommended operating range from: T_A = -40°C to +85°C, VCC= 2.7V to 3.6V, unless otherwise noted.

(2) Typical program time is defined as the time within which more than 50% of the whole pages are programmed at 3.3V and 25°C.

(3) These parameters are characterized only.

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【13 时序图】

图28 串行输入时序：标示 CS#、CLK、SIO INPUT 的 tCHSL、tSLCH、tDVCH、tCHDX、tCLCH、tCHCL、tCHSH、tSHCH、tSHSL 等关系，数据按 MSB IN 到 LSB IN 输入。

图29 串行输出时序：标示 tCLQV、tCLQX、tCLH、tCLL、tSHQZ，最后输出 LSB OUT。

图30 HOLD# 时序：HOLD#/SIO3 与 CLK、CS#、SIO 输出/输入之间的 tCHHL、tHLCH、tCHHH、tHHCH、tHLQZ、tHHQX 关系。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



13 Timing Diagram

Figure 28 Serial Input Timing

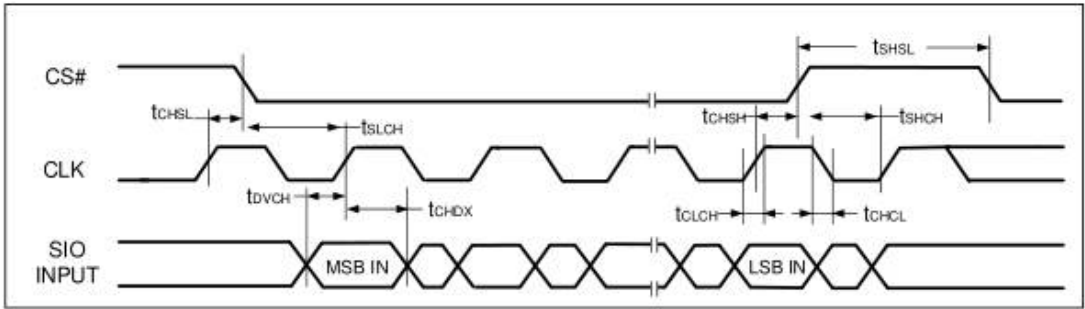


Figure 29 Serial Output Timing

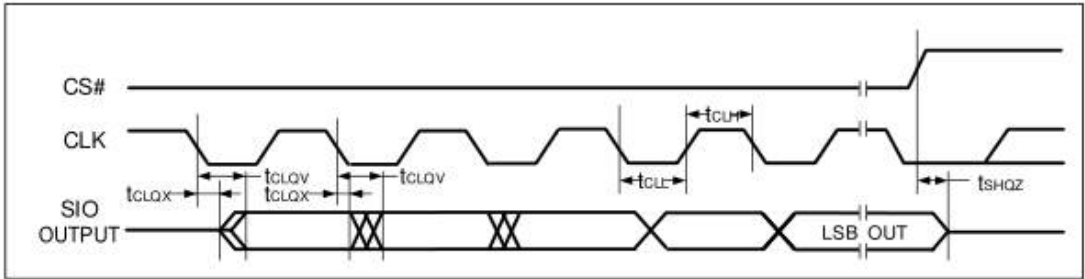
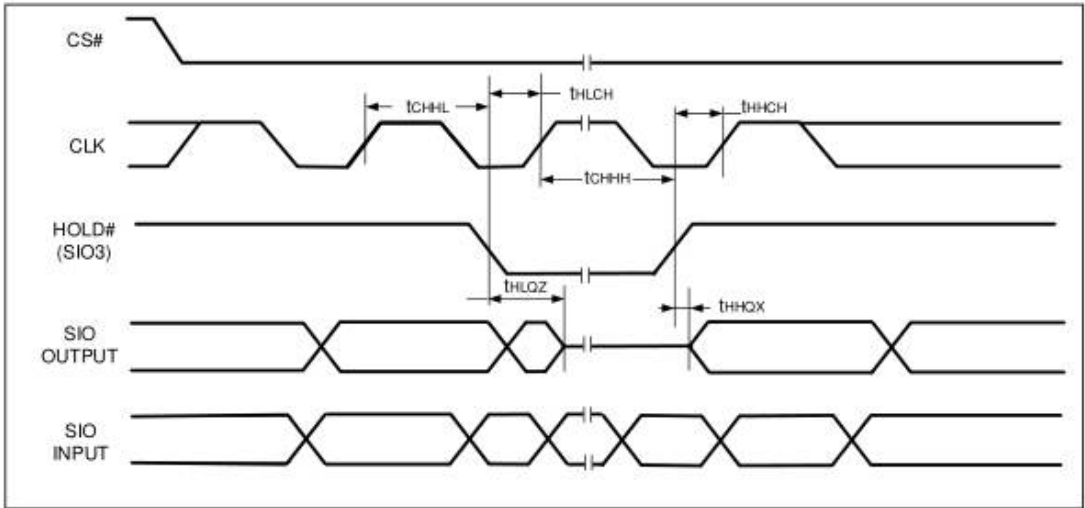


Figure 30 HOLD# Timing



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【图31 WP# 时序】

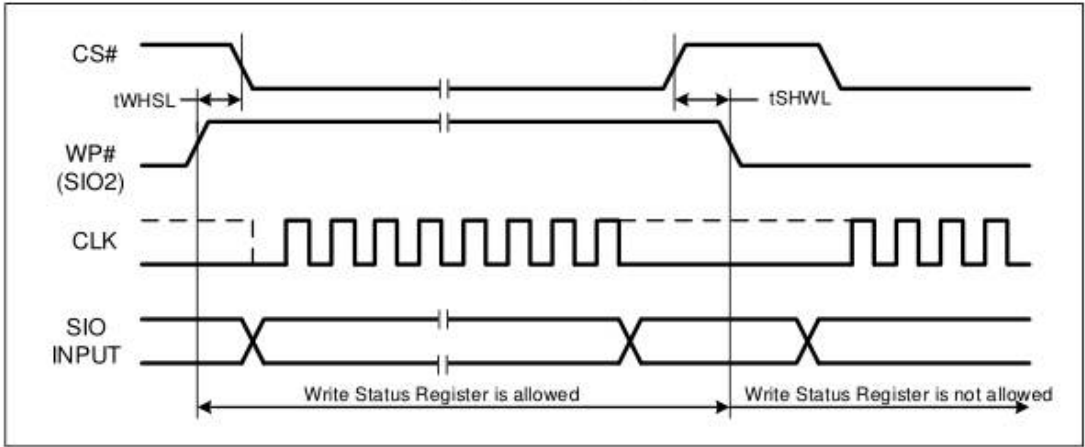
WP#/SIO2 在 CS# 拉低前须满足 tWHSL setup，在 CS# 拉低后保持 tSHWL。图中左侧阶段允许写 Status/Protection Register，进入受保护阶段后不允许写。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Figure 31 WP# Timing



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【14 器件顶标规则】

14.1 8-WSON (8x6mm)

典型顶标：

FORESEE

F35SQB004G

XXXXXXXXXX

YYWWTT

其中：Trace code = 追溯码；WW = 产品塑封时的生产周（例如第32周）；YY = 产品塑封年份末两位；Assembly lot number = 封装装配批号。

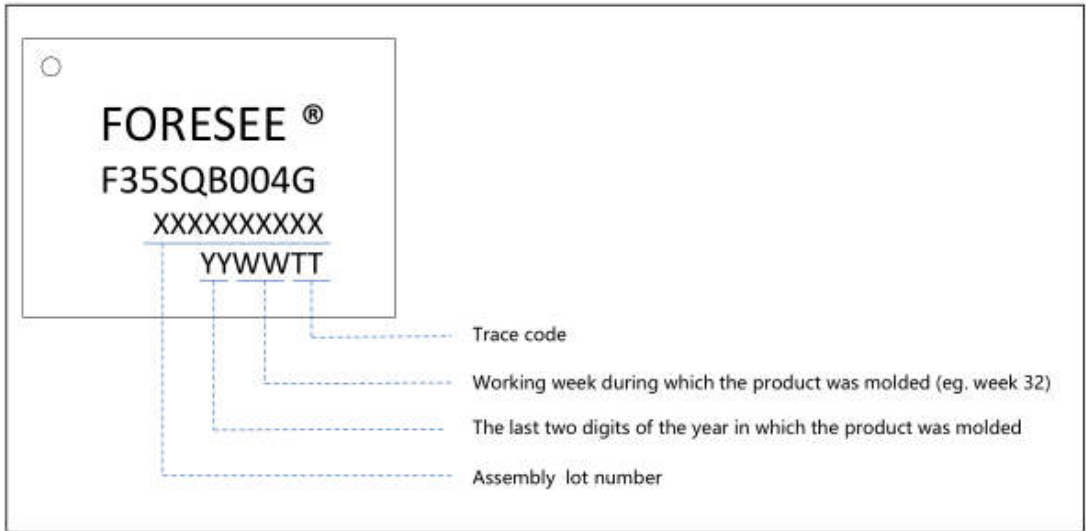
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



14 Part Marking Scheme

14.1 8-WSON (8x6mm)



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【15 封装信息】

15.1 8-WSON (8x6mm) 封装尺寸 (单位 mm)

A : 0.70 / 0.75 / 0.80 (Min/Nom/Max)

A1 : 0.00 / 0.02 / 0.05

A3 : 0.20 REF

D : 5.90 / 6.00 / 6.10

E : 7.90 / 8.00 / 8.10

b : 0.35 / 0.40 / 0.45

D2 : 4.20 / 4.30 / 4.40

E2 : 3.30 / 3.40 / 3.50

e : 1.27 BSC

L : 0.45 / 0.50 / 0.55

详细机械轮廓、焊盘位置与尺寸公差见原页封装图。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



15 Packaging Information

15.1 8-WSON (8x6mm)

Figure 32 8-WSON (8x6mm) Package Information

